LOVELY PROFESSIONAL UNIVERSITY
Transforming Education Transforming India

NAAC GRADE A++

Division of Academic Affairs

Recognized by UGC under Section 2(f) of the University Grants Commission Act, 1956 by the Notification No. F.9-10/ (CPP-1)

Member of Association of Indian Universities (AIU), New Delhi

LPU/10.1/ARC1(CURR)/2024/005(5.4)/EC/260410/0001

Date: 10-Apr-26

TO WHOM SO EVER IT MAY CONCERN

Name of the Student	Mr. Anmol Raj
Son of	Mr. Rajesh Kumar
Registration No.	11908117
Name of the Programme	Bachelor of Technology (Computer Science and Engineering)
Batch	2019
Faculty	Lovely Faculty of Technology & Science

This is to certify that the student has completed above said programme at the university in the year 2023. The certified copy of the curriculum applicable to batch specified above is enclosed.

Dean
Division of Academic Affairs

Dean
Division of Academic Affairs
Lovely Professional University

Prepared by

Verified by

Checked by

Disclaimer: This certificate is issued on the basis of information available in the office of records on the date of its issue and the University reserves the right to update/change any information contained herein without notice further the University expressly disclaims all obligations to confirm the accuracy of any of the particulars in this certificate based upon information submitted by the candidate.

CSE101: COMPUTER PROGRAMMING

L: 2 T: 0 P: 2 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: discuss the various approaches towards solving a particular problem using the C language constructs

CO2 :: write programs to solve different problems using C constructs irrespective of the compilers

CO3 :: plan the process of code reuse by forming a custom library of one's own functions

CO4 :: complete the understanding and usage of one of the building blocks of data structures namely pointers

CO5 :: categorize the theoretical knowledge and insights gained thus far to formulate working code

CO6 :: validate the underlying logic and formulate code which is capable of passing various test cases

UNIT-I

Basics and introduction to C: The C character set, Identifiers and keywords, Data types, Constants and variables, Expressions, Arithmetic operators, Unary, Relational, Logical, Assignment and conditional operators Bitwise operators

UNIT-II

Control structures and Input/output functions: If, If else, Switch case statements, While, For, Do-while loops, Break and continue statements, Go to, Return, Type conversion and type modifiers, Designing structured programs in C, Formatted and unformatted Input/output functions like print f(), Scan f(), Puts(), Gets() etc.

UNIT-III

User defined functions and Storage classes: Function prototypes, Function definition, Function call including passing arguments by value and passing arguments by reference, Math library functions, Recursive functions, Scope rules (local and global scope), Storage classes in C namely auto Extern, Register, Static storage classes

UNIT-IV

Arrays in C: Declaring and initializing arrays in C, Defining and processing 1D and 2D arrays, Array applications, Passing arrays to functions, inserting and deleting elements of an array, Searching including linear and binary search methods, Sorting of array using bubble sort

UNIT-V

Pointers, Dynamic memory allocation: Pointer declaration and initialization, Types of pointers - dangling , wild, null, generic (void), Pointer expressions and arithmetic, Pointer operators, Operations on pointers, Passing pointer to a function, Pointer and one dimensional array, Dynamic memory management functions (malloc, calloc, realloc and free)

Assistant Dean
Division of Academic Affairs
Lovely Professional University

UNIT-VI

Strings, Derived types including structures and unions: Defining and initializing strings, Reading and writing a string, Processing of string, Character arithmetic, String manipulation functions and library functions of string, Declaration of a structure, Definition and initialization of structures, Accessing structures, Structures and pointers, Nested structures, Declaration of a union, Macros and its types(object-like function-like chain multi-line)

LABORATORY WORK:

- Basics and introduction to C: Programs to explore different data types and usage, Programs for different type of operators and the usage
- Control structures and Input/output functions: Programs on decision making constructs as if, if else and switch Programs on formatted and unformatted functions as print f() scan f() gets() and puts()
- User defined functions, Storage classes: Program to explore different prototypes, Program to differentiate between call by value, call by address, Program to demonstrate storage classes as auto, register extern static
- Arrays in C: Program to demonstrate memory arrangement of 1D and 2D array, Program to demonstrate operations on array as insertion, deletion, searching (linear, binary),Program to demonstrate bubble sort
- Pointers, Dynamic memory allocation: Program to demonstrate type of pointers, Program to demonstrate pointer vs. array name, Program to demonstrate dynamic memory management functions (malloc(), calloc(), realloc() and free()
- Strings, User defined types including structures and unions: Program to demonstrate string operations, Program to demonstrate structure and nested structures, Program to differentiate between structure and union

RECOMMENDED READINGS:

1. PROGRAMMING IN C BY ASHOK N. KAMTHANE, PEARSON EDUCATION INDIA.
2. PROGRAMMING IN ANSI C BY E. BALAGURUSAMY, TATA MCGRAW HILL, INDIA.
3. C HOW TO PROGRAM BY PAUL DEITEL AND HARVEY DEITEL, PEARSON EDUCATION INDIA.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

ECE131: BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

L: 3 T: 1 P: 0 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: analyze DC circuits and conceive approaches in developing and designing of DC circuits.

CO2 :: analyze AC Circuits and develop solutions to problems and conceive approaches in developing and designing of AC circuits.

CO3 :: develop the knowledge about the characteristics and working principles of transformers and motors.

CO4 :: develop basic knowledge of Laplace transform and its application in circuit analysis.

CO5 :: develop basic knowledge in operational amplifier and its application as filters.

CO6 :: develop basic knowledge of embedded system and its application in industry.

UNIT-I

Fundamentals of D.C. circuits: use of scientific calculators, the circuit abstraction - power of abstraction, lumped circuit abstraction, lumped matter discipline, limitations of the lumped circuit abstraction, practical two-terminal elements, ideal two-terminal elements, resistive networks-current division rule, voltage division rule, star–delta transformation, Kirchhoff's laws, basic method of circuit analysis, intuitive method of circuit analysis- series and parallel simplification, ideal and non-ideal energy sources, dependent sources and the control concept, network theorems-node method, loop method, superposition theorem, Thevenin's theorem, Norton's theorem, maximum power transfer theorem

UNIT-II

Fundamentals of A.C. circuits: alternating current and voltage, concept of notations (i, v, I, V), three-phase circuits- numbering and interconnection (delta or mesh connection) of three phases, relations in line and phase voltages and currents in star and delta, series and parallel connections, sinusoidal inputs, step inputs, impulse inputs, use of Laplace transform in circuit analysis, first-order transients in linear electrical networks - analysis of RC circuits, analysis of RL circuits, impedance and frequency response - series RL circuit, RC circuit, frequency response of capacitors, inductors and resistors, power and energy in impedance, pure resistance, pure reactance, frequency response for resonant systems, series RLC, reactive power and power calculations, methods of power factor correction

UNIT-III

Fundamentals of electrical machines: Fleming's left hand and right hand rule, mutual inductance and mutual coupling phenomena in transformer, transformer – working, concept of turns ratio and applications, transformer on DC, auto-transformer, instrument transformers, dc machines- working principles, classification, starting, speed control and applications of dc motors, working principle of single and three phase induction motors, applications of ac motors

Assistant Dean
Division of Academic Affairs
Lovely Professional University

UNIT-IV

Fundamentals of semiconductor devices: digital abstraction- voltage levels and the static discipline, Boolean logic, combinational gates, fan-in and fan-out of gates, noise margin in details, diodes- semiconductor diode characteristics, analysis of diode circuits, clamping circuit, testing of diodes, basic operation and testing of BJT, MOSFET-representation and characteristics, handling of integrated circuits-ESD phenomena

UNIT-V

Fundamentals of filters and operational amplifier: filter examples- band-pass filter, low-pass filter, high-pass filter, operational amplifier abstraction- device properties of the operational amplifier, simple op amp circuits – virtual ground concept, inverting and non-inverting op-amp, op-amp as an adder and subtractor, op-amp RC circuits – op-amp integrator, op-amp differentiator, active filter, op- amp as a comparator and its application in anti-lock braking systems

UNIT-VI

Fundamentals of embedded system and its application in industrial processes: comparison of microprocessor and micro-controller, types of processors: SOC, ASIC, DSP and FPGA, introduction to embedded system, examples of real-time applications of embedded system: GPOS and RTOS, cyber physical world, role of IOT and cloud computing in condition monitoring of plant processes

RECOMMENDED READINGS:

1. FOUNDATIONS OF ANALOG AND DIGITAL ELECTRONIC CIRCUITS BY ANANT AGARWAL AND JEFFREY H.LANG, MORGAN KAUFMANN.
2. INTRODUCTION TO ELECTRONICS BY EARL GATES, DELMAR - CENGAGE LEARNING.
3. BASIC ELECTRICAL ENGINEERING BY D.C. KULSHRESTHA, MC GRAW HILL.
4. INTRODUCTION TO EMBEDDED SYSTEMS BY K. V. SHIBU, MC GRAW HILL.

Singh

Assistant Dean
Division of Academic Affairs
Lovely Professional University

ECE132: BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY

L: 0 T: 0 P: 2 Credits: 1

COURSE OUTCOMES: Through this course students should be able to

CO1 :: develop circuit models for elementary electronic components like resistors, sources, inductors, capacitors, diodes and transistors

CO2 :: apply the knowledge about semiconductors devices and basic digital gates in circuit designing

CO3 :: employ the knowledge of math, science and engineering while implementing and analyzing electrical and electronics engineering problems

LABORATORY WORK:

Kirchhoff voltage law and Kirchhoff current law

- Verification of Kirchhoff's voltage law and Kirchhoff's current law

Turn ratio of a transformer

- To understand the principle of turn ratio of a transformer.

Thevenin's and Norton's theorems

- Verification of Thevenin's and Norton's theorems in DC circuits along with simulation on P-Spice.

Comparison of different lighting sources

- To compare incandescent lamp, fluorescent lamp, CFL and LED based light source for its efficiency.
- Switching control of single lamp by using four 2 way switches.

Distribution Board

- To learn the use of electrical fuse, MCB, energy meter, house wiring and connections of switches.

Rectifiers

- To understand use of diodes for half wave and full wave rectifiers

DC Motors

- To understand principle of speed control of a DC motor using hardware and Proteus software.

Low pass filter and high pass filter

- To study the effect of frequency on the output voltage in low-pass and high-pass filters.

Logic gates and verification of Boolean expression

- To understand Truth table of Logic Gates and verifying Boolean equations

Zener diode characteristics

- To study VI char of a Zener diode and its application as a voltage regulator.

RECOMMENDED READINGS:

1. BASIC ELECTRICAL & ELECTRONICS BY B.L THARAJA, S. CHAND & COMPANY.
2. MICROELECTRONICS CIRCUITS: THEORY AND APPLICATIONS BY ADEL S. SEDRA, OXFORD & IBH.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

MEC103: ENGINEERING GRAPHICS

L: 2 T: 2 P: 0 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: visualize the knowledge of basic geometries, geometric tools, shapes and procedures used in engineering drawing.

CO2 :: represent detailed conceptual knowledge about the dimensioning, specifications and conventions.

CO3 :: understand the concept of projection and acquire visualization skills, projection of points.

CO4 :: understand the concept to draw the basic views related to projections of Lines.

CO5 :: understand different concepts of sectioning and 3-D representations of objects.

CO6 :: sketch the different concepts of isometric projections

CO7 :: understand the different concepts of development of surfaces.

CO8 :: develop the visualization skills to apply these skill in the development of new products.

UNIT-I

Introduction to Engineering Drawing: Principles of Engineering Graphics and their significance, Drawing instruments, Lettering in vertical Gothic letters using single stoke, Dimensioning, Different types of lines used in engineering drawing, Plane and Diagonal Scale, Basics of BIS norms

UNIT-II

Projection of Points and Lines: Projection of Points, Projection of line perpendicular to HP, Projection of line perpendicular to VP, Projection of line parallel to HP and VP, Projection of line inclined to HP and parallel to VP, Projection of line inclined to VP and parallel to HP, Horizontal and Vertical traces of line

UNIT-III

Orthographic Projections: Method of obtaining Orthographic Projections in First angle and third angle projection. Principles of orthographic projections

UNIT-IV

Sectional views: Sectioning webs and ribs, Importance of sectioning, Types of section including full section, offset section and half section

UNIT-V

Isometric Projections: Principles of Isometric Projections, Isometric Scale, Terminology, Isometric view of step, inclined, oblique, cylindrical blocks, Isometric Dimensioning

UNIT-VI

Development of Surfaces: Methods of development, Parallel line development of cylinder and prism, Radial line development of cone and pyramid

RECOMMENDED READINGS:

1. ENGINEERING DRAWING WITH AN INTRODUCTION TO AUTOCAD BY DHANANJAY A JOLHE, MCGRAW HILL EDUCATION
2. ENGINEERING GRAPHICS BY AMAR PHATAK BY AMAR PHATAK, DREAMTECH PRESS
3. ENGINEERING DRAWING BY M.B.SHAH,BC RANA, PEARSON

Assistant Dean
Division of Academic Affairs
Lovely Professional University

4. ENGINEERING GRAPHICS BY K C JOHN, PRENTICE HALL
5. ENGINEERING DRAWING BY N.D. BHAT & M. PANCHAL, CHAROTAR PUBLISHING HOUSE PVT. LTD.
6. ENGINEERING DRAWING AND DESIGN BY JENSEN, HELSEL AND SHORT, MCGRAW HILL EDUCATION

Assistant Dean
Division of Academic Affairs
Lovely Professional University

MTH165: MATHEMATICS FOR ENGINEERS

L: 3 T: 1 P: 0 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: recall the concepts of matrices and its application to solve the system of linear equations.

CO2 :: review the basic concept of calculus of one variable.

CO3 :: apply the concept of calculus to evaluate extreme values of functions.

CO4 :: describe calculus of multivariate functions and their applications.

CO5 :: evaluate surface and volume integral using multiple integral.

CO6 :: describe the concept of Fourier series and its application.

UNIT-I

Linear Algebra: Review of matrices, Elementary operations of matrices, Rank of a matrix, Linear dependence and independence of vectors, Solution of Linear system of equations, Inverse of matrices Eigen values and Eigen vectors, Properties of Eigen values, Cayley-Hamilton theorem

UNIT-II

Differential and integral calculus: General rules of differentiation, Derivatives of standard functions, Derivatives of Parametric forms, Derivatives of implicit functions, Logarithmic differentiation,, properties of indefinite integral, Methods of integration-By Parts, Methods of integration-By Partial fractions Properties of definite integral

UNIT-III

Application of derivatives: Role's theorem, Mean value theorems, Taylor's theorems with remainders, Maclaurin theorems with remainders, indeterminate forms, L' Hospital's rule, maxima and minima.

UNIT-IV

Multivariate functions: Functions of two variables, Limits and Continuity, Partial derivatives, Total derivative and differentiability, Chain rule, Euler's theorem for Homogeneous functions, Maxima and Minima, Lagrange method of multiplier

UNIT-V

Multiple Integrals: Double integrals, change of order of integration, Triple integrals, change of variables, Application of double integrals to calculate area and volume, Application of triple integrals to calculate volume.

UNIT-VI

Fourier series: Introduction and Euler's formulae, Conditions for a Fourier Expansion and Functions having points of discontinuity, Change of interval, Even and odd functions, Half Range series, Perseval's Formula, Complex form of Fourier Series

RECOMMENDED READINGS:

1. ADVANCED ENGINEERING MATHEMATICS BY R.K.JAIN, S. R. K. IYENGER AND NAROSA PUBLISHING HOUSE.
2. HIGHER ENGINEERING MATHEMATICS BY B. S. GREWAL, KHANNA PUBLISHERS.
3. MATHEMATICS TEXT BOOKS FOR CLASS XII PART I BY NCERT.
4. MATHEMATICS TEXT BOOK FOR CLASS XII PART II BY NCERT.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

PES318: SOFT SKILLS-I

L: 1 T: 2 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: demonstrate optimism to develop positive attitude

CO2 :: articulate fluently with confidence

CO3 :: illustrate persuasive and negotiation skills

CO4 :: develop skills to meet the industry expectation

CO5 :: understand the importance of corporate practices

UNIT-I

Attitude Reconstruction: understanding emotional intelligence and exercising in professional life, building self-esteem and self-confidence, types of personalities, empathy, understanding mission, vision and career

UNIT-II

Mastering Communication Skills: introduction to communication process, types of communication; verbal communication (speaking and listening), importance of verbal communication, significance of non-verbal communication; intonation, body language, communication in organizations communication barriers

UNIT-III

Personal and Social Branding: introduction to personal branding, view yourself as a brand, social media strategy, offline branding, resume, digital profiling, understanding the concept of unique selling points power dressing

UNIT-IV

Group Discussion: introduction to group discussions, types of group discussion topics, advantages and conclusion of group discussions, do's and don'ts of group discussions, ideation techniques- keyword analysis, social, political, economic, legal and technology technique, hurdle up, open house discussions

UNIT-V

Interview Skills: professional grooming, know your company, types of interviews, interview answering techniques, case studies

UNIT-VI

Workplace Etiquette: e-mail etiquette, small talk, building a rapport, elevator pitch, leadership and teamwork, art of giving and receiving feedback

RECOMMENDED READINGS:

1. SOFT SKILLS: KNOW YOURSELF AND KNOW THE WORLD BY DR. K. ALEX, S CHAND PUBLISHING.
2. PERSONALITY DEVELOPMENT AND SOFT SKILLS BY BARUN K. MITRA, OXFORD UNIVERSITY PRESS.
3. THE ACE OF SOFT SKILLS: ATTITUDE, COMMUNICATION AND ETIQUETTE FOR SUCCESS BY GOPALASWAMY RAMESH AND MAHADEVAN RAMESH, PEARSON.

PHY109: ENGINEERING PHYSICS

L: 3 T: 1 P: 0 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: recall the basic principles of physics in engineering courses

CO2 :: extend the knowledge on electromagnetic theory

CO3 :: articulate the concepts of laser and its application

CO4 :: discover the concepts of physics in understanding fibre optics

CO5 :: analyze the importance of quantum physics and its application

CO6 :: evaluate the need of ultrasonic waves and its generation mechanism

CO7 :: develop the concepts of solid state physics

CO8 :: relate physical understanding with Engineering design

UNIT-I

Electromagnetic theory: scalar and vectors fields, concept of gradient, divergence and curl, dielectric constant, Gauss theorem and Stokes theorem (qualitative), Poisson and Laplace equations, continuity equation, Maxwell electromagnetic equations (differential and integral forms), physical significance of Maxwell equations, Ampere Circuital Law, Maxwell displacement current and correction in Ampere Circuital Law.

UNIT-II

Lasers and applications: fundamentals of laser- energy levels in atoms, Radiation matter interaction, Absorption of light, spontaneous emission of light, stimulated emission of light, population of energy levels, Einstein A and B coefficients, metastable state, population inversion, lasing action, properties of laser, resonant cavity, excitation mechanisms, Nd - YAG, He-Ne Laser, Semiconductor Laser, applications of laser in engineering, holography.

UNIT-III

Fiber optics: fiber optics introduction, optical fiber as a dielectric wave guide, total internal reflection, acceptance angle, numerical aperture, relative refractive index, V-number, step index and graded index fibers, losses associated with optical fibers, application of optical fibers.

UNIT-IV

Quantum mechanics: need of quantum mechanics, photoelectric effect, concept of de Broglie matter waves, wavelength of matter waves in different forms, Heisenberg uncertainty principle, concept of phase velocity and group velocity (qualitative), wave function and its significance, Schrodinger time dependent and independent equation, particle in a box.

UNIT-V

Waves: interference phenomenon, concept of resonance, production of ultrasonic waves by magnetostriction method, audible, ultrasonic and infrasonic waves, production of ultrasonic waves by piezoelectric method, ultrasonic transducers and their uses, applications of ultrasonic waves, detection of ultrasonic waves (Kundt's tube method, sensitive flame method and piezoelectric detectors), absorption and dispersion of ultrasonic waves.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

UNIT-VI

Solid state physics: free electron theory (Introduction), diffusion and drift current (qualitative)., Fermi energy, Fermi-Dirac distribution function, band theory of solids - formation of allowed and forbidden energy bands, concept of effective mass - electrons and holes, Hall effect (with derivation), semiconductors and insulators, Fermi level for intrinsic and extrinsic semiconductors, direct and indirect band gap semiconductors.

RECOMMENDED READINGS:

1. ENGINEERING PHYSICS BY HITENDRA K MALIK, A K SINGH, MC GRAW HILL.
2. ENGINEERING PHYSICS BY B K PANDEY, S CHATURVEDI AND CENGAGE LEAR.
3. ENGINEERING PHYSICS BY D K BHATTACHARYA, POONAM TONDON AND OXFORD UNIVERSITY PRESS.
4. FUNDAMENTALS OF PHYSICS BY HALLIDAY, RESNICK, WALKER, WILEY.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

PHY119: ENGINEERING PHYSICS LABORATORY

L: 0 T: 0 P: 2 Credits: 1

COURSE OUTCOMES: Through this course students should be able to

CO1 :: associate practical knowledge with the theoretical studies.

CO2 :: demonstrate basic experimental techniques required to find fundamental parameters in physics.

CO3 :: develop good experimental skills, including proper setup, care of equipment, conducting experiments and analyzing results in order to observe physical phenomena.

LABORATORY WORK:

- To find the wavelength of sodium light by measuring the diameter of Newton rings.
- To investigate the intensity of light coming through two crossed polaroids and to verify the Malus law.
- To determine Hall voltage and Hall coefficient using Hall effect and also find carrier concentration of the crystal used.
- To determine the dielectric constant of solid by resonance method.
- To find out the energy band gap of semiconductor using four probes method.
- To study the cell.
- To plot a graph between current and frequency in series and parallel LCR circuit and to find the resonant frequency, quality factor and bandwidth.
- To find the frequency of AC mains variation of magnetic field with the distance along the axis of circular coil carrying current by plotting a graph and also to find the radius of the circular coil.
- To determine the velocity of ultrasonic waves using ultrasonic interferometer and to find the compressibility of the given liquid.
- To find the value of Planck's constant and photoelectric work function of the material of the cathode using a photoelectric using electric vibrator.

RECOMMENDED READINGS:

1. B.SC. PRACTICAL PHYSICS BY HARNAM SINGH, DR P.S. HEMNE, S. CHAND PUBLISHING.
2. B.SC. PRACTICAL PHYSICS BY C.L. ARORA, S. CHAND & COMPANY.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CHE110: ENVIRONMENTAL STUDIES

L: 2 T: 2 P: 0 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: understand the current environmental issues and associated problems.

CO2 :: gain the basic knowledge of environment and its various components.

CO3 :: spread the environmental awareness among people.

UNIT-I

Introduction and natural resources: Multidisciplinary nature of environmental studies, Scope and importance: Concept of sustainability and sustainable development, Land resources: Land degradation, soil erosion and desertification, Deforestation: Causes and impacts due to mining, dam building on environment, forests, biodiversity and tribal populations, Water: Use and over-exploitation of surface and ground water, floods, droughts, conflicts over water, Energy resources: Renewable and non renewable energy sources, use of alternate energy sources, growing energy needs, case studies.

UNIT-II

Ecosystems: What is an ecosystem? Structure and function of ecosystem, Energy flow in an ecosystem: food chains, food webs and ecological succession, ecological pyramids, Case studies of the following ecosystems: a)forest ecosystem b) grassland ecosystem c) desert ecosystem d) aquatic ecosystem.

UNIT-III

Biodiversity and conservation: Levels of biological diversity: genetic, species and ecosystem diversity, Biogeographic zones of India, Biodiversity patterns and global biodiversity hot spots, India as a mega diversity nation, Endangered and endemic species in India, Threats to biodiversity: Habitat loss, poaching of wildlife, man-wildlife conflicts, biological invasions, Conservation of biodiversity: In- situ and ex-situ conservation of biodiversity, Ecosystem and biodiversity services: ecological, economic, social, ethical, aesthetic and Informational value.

UNIT-IV

Environmental pollution: Environmental pollution: types, causes, effects and controls; Air, Ill- effects of Fireworks, water, soil and noise pollution, Nuclear hazards and human health risks, Pollution case studies, ill-effects of Fireworks.

UNIT-V

Environmental Policies & Practices: Climate change, global warming, ozone layer depletion, acid rain and impacts on human communities and agriculture, Environment Laws: Environment Protection Act, Air (Prevention & Control of Pollution) Act, Water (Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act, International agreements: Montreal and Kyoto protocols and Convention on Biological Diversity (CBD), Nature reserves, tribal populations and rights, and human wildlife conflicts in Indian context, Solid waste management: Control measures of urban and industrial waste.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

UNIT-VI

Human Communities and the Environment: Human population growth: Impacts on environment, human health and welfare, Disaster management: floods, earthquake, cyclones and landslides, Environmental movements: Chipko, Silent valley, Bishnois of Rajasthan, Environmental ethics: Role of Indian and other religions and cultures in environmental conservation, Environmental communication and public awareness, case studies.

RECOMMENDED READINGS:

1. PERSPECTIVE IN ENVIRONMENTAL STUDIES BY ANUBHA KAUSHIK, C P KAUSHIK AND NEW AGE INTERNATIONAL PUBLISHERS.
2. TEXT BOOK OF ENVIRONMENTAL STUDIES BY D. DAVE AND S. S. KATEWA, CENGAGE LEARNING.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE202: OBJECT ORIENTED PROGRAMMING

L: 3 T: 0 P: 2 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: identify basic programming constructs and use the newly acquired skills to solve extensive programming problems

CO2 :: discuss the mechanism of code reusability by creating own libraries of functions

CO3 :: validate the logic building and code formulation by designing code capable of passing various test cases

CO4 :: interpret the principles of the object-oriented model and apply it in the implementation in C ++ language

CO5 :: develop accurate, reliable and efficient software applications

CO6 :: apply the knowledge acquired to develop software applications

UNIT-I

Concepts and Basics of C++ Programming: Differences between procedural and object oriented programming paradigms, Reading and writing data using cin and cout, Creating classes, Class objects, Accessing class members, Differences between Structures, Unions and Classes, Inline and Non-inline member functions, Static data members and static member functions.

Functions and Input/output Streams: Functions with Default parameters/arguments, Inline Functions, Features of Input/output Streams, Manipulators Functions, Function overloading and Scope rules, Friend of a class (friend function and friend class), Reference variables, Differences between Call by value, Call by address and call by reference, Recursion

UNIT-II

Pointers, Reference Variables, Arrays and String Concepts: Differences between pointer and reference variables, Void pointer, Pointer arithmetic, Pointer to pointer, Possible problems with the use of pointers - Dangling pointer, Wild pointer, Null pointer assignment, Classes containing pointers, Pointer to objects, this pointer, Pointer to a member, Array declaration and processing of multidimensional arrays, Array of objects, The Standard C++ string class-defining and assigning string objects, Member functions, Modifiers of string class

UNIT-III

Constructors, Destructors and File Handling: Manager functions (constructors and destructor), Default constructor, Parameterized constructor, Copy constructor, Initializer lists, Constructor with default arguments, Destructors.

Data File operations: Opening and closing of files, Modes of file, File stream functions, Reading/Writing of files, Sequential access and random access file processing, Binary file operations, Classes and file operations, Structures and file operation

UNIT-IV

Operator Overloading and Type Conversion : Operator Overloading (unary operator, binary operator overloading), Type conversions - basic type to class type, class type to basic type

Inheritance: Inheritance Basics – derived class and base class, Types (simple, multi-level, multiple and hierarchical), Modes (private, protected, public inheritance), Overriding member functions, Order of execution of constructors and destructors, Resolving ambiguities in inheritance, Virtual base class

UNIT-V

Dynamic Memory Management and Polymorphism : Dynamic memory allocation using new and delete operators, Memory leak and allocation failures, Virtual destructors, Compile and run time polymorphism, Virtual functions, Pure virtual functions, Abstract classes, Early binding and late binding, Dynamic constructors

UNIT-VI

Exception Handling, Templates and Standard Template Library (STL) : Basics of exception handling, Exception handling mechanism, Throwing mechanism, Catching mechanism, Re-throwing an exception, Function template and class template, Introduction to STL- Containers, Algorithms and iterators, Container - Vector and List

LABORATORY WORK:

Concepts and Basics of C++ Programming

- Programs to define classes and structures.
- Program to demonstrate inline, non inline member functions and Static function

Functions and Input/output Streams

- Program to implement function overloading, friend function and friend class.
- Program to demonstrate the difference between call by value, call by address and call by reference.

Pointers, Reference Variables, Arrays and String Concepts

- Program to demonstrate the type of pointers.
- Program to process multidimensional array and array of objects.

Constructors, Destructors and File Handling

- Program to demonstrate constructor, destructor and type of constructors.

Data File operations

- Program to demonstrate the modes of file.
- Program to demonstrate type of files.

Operator Overloading and Type Conversion

- Program to demonstrate the operator overloading and type conversion.

Inheritance

- Program to demonstrate the type of inheritance.
- Program to demonstrate the ambiguities in inheritance.

Dynamic Memory Management and Polymorphism

- Program to use new and delete for dynamic memory management.
- Program to demonstrate the compile time and run time polymorphism.
- Program to demonstrate abstract class and dynamic constructor.

Exception Handling, Templates and Standard Template Library (STL)

- Program to demonstrate exception handling.
- Program to demonstrate function template and class template.
- Program to demonstrate STL- Containers, Algorithms and Iterators

RECOMMENDED READINGS:

1. OBJECT ORIENTED PROGRAMMING IN C++ BY ROBERT LAFORE, PEARSON
2. PROGRAMMING WITH C++ BY D RAVICHANDRAN, MCGRAW HILL EDUCATION
3. OBJECT ORIENTED PROGRAMMING IN C++ BY E BALAGURUSAMY, MCGRAW HILL EDUCATION

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE326: INTERNET PROGRAMMING LABORATORY

L: 0 T: 0 P: 2 Credits: 1

COURSE OUTCOMES: Through this course students should be able to

CO1 :: understand the principles of creating an effective web page, including an in-depth consideration of information architecture.

CO2 :: develop a web page using HTML and CSS.

CO3 :: develop responsive web pages with the help of CSS and Java script.

CO4 :: demonstrate the JavaScript functions and events.

CO5 :: construct websites using text, images, links, lists, and tables for navigation and layout.

LABORATORY WORK:

Exposure To HTML

- HTML document structure
- Working with HTML basic elements like title, head, body
- Working with Root and Metadata
- Script and Non Script
- Horizontal Rules and line breaks and paragraph, working with citation, quotation, definitions and comments
- Types of Tags in HTML
- Working with Text, Links, Images , URLS, Multimedia and Interactive in HTML
- Formatting text with HTML physical style elements, Formatting text with HTML logical style elements
- Creating links with anchor tag
- Multimedia- audio, tags and attributes like controls, auto play and loop
- Working with images in a web page

Cascading Style Sheets

- Introduction To CSS, Inline CSS, internal CSS and external CSS
- CSS selectors-type, id and class
- CSS properties-text controlling and text formatting
- CSS Box Model- Padding, Margin, Border
- Div and Span Tag in CSS, Working with background Images

Working with Tables

- Working With Tables-Cols pan and Row span
- applying css on tables
- creating hover able tables

Working with Forms

- Working with Forms- action attribute, get and post methods
- Form Elements and Controls Like Text Inputs, , Buttons, Check Boxes, Dropdown Boxes, Radio Buttons Select and File Select
- applying css on all controls of forms

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSS Grids and Webpage Layouts

- Grid introduction
- Grid container
- Grid Item
- Creating different layouts for webpage
- JavaScript Application Development
- Incorporating JavaScript in the HEAD and BODY element, Using an External JavaScript file, Using variables and operators
- Using control statements such as if...else, switch, break and continue, Using looping statements such as while, do...while, for, Using Popup boxes such as Alert, Confirm, and Prompt Working with JavaScript Objects ,Properties and Methods

JavaScript Functions, Events and Validation

- Working with Functions-Using function arguments and return statement
- Working with JavaScript Events like Form Based, Keyboard Based and Mouse Based
- JavaScript Form Validation

Java script DOM

- DOM introduction
- DOM methods
- DOM document
- DOM elements
- DOM HTML, DOM CSS
- DOM Events

RECOMMENDED READINGS:

1. HTML 5 COVERS CSS3, JAVASCRIPT,XML,XHTML,AJAX BY KOGENT LEARNING, DREAMTECH PRESS
2. WEB ENABLED COMMERCIAL APPLICATION DEVELOPMENT USING HTML,
3. BEGINNING HTML, XHTML, CSS AND JAVASCRIPT BY JON DICKETT, WILEY

Assistant Dean
Division of Academic Affairs
Lovely Professional University

ECE213: DIGITAL ELECTRONICS

L: 3 T: 1 P: 0 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: recall the basics of digital electronic logics and circuits.

CO2 :: apply Boolean algebra in minimize the complex Boolean expression

CO3 :: distinguish between combinational logic system and sequential logic system

CO4 :: analyze the basic techniques in designing and implementing digital systems.

UNIT-I

Number Systems : Digital Systems, Data representation and coding, Logic circuits, Implementation of digital systems, Number Systems, Codes- Positional number system, Binary number system, Methods of base conversions, Binary arithmetic, Representation of signed numbers, Fixed numbers, Binary coded decimal codes, Gray codes, Error detection code, Parity check codes, octal number system, Hexadecimal number system, Error correction code, Hamming code, Octal arithmetic, Hexadecimal arithmetic Floating point numbers

UNIT-II

Combinational Logic System : Truth table, Basic logic operation, Boolean Algebra, Basic postulates, Standard representation of logic functions -SOP forms, Simplification of switching functions - K-map, Synthesis of combinational logic circuits, Logic gates, Fundamental theorems of Boolean algebra Standard representation of logic functions POS forms

UNIT-III

Introduction to Combinational Logic Circuits : Adders, Subtractors, Comparators, Multiplexers and De-multiplexers, Decoders, Encoders, Parity circuits

Introduction to Logic Families : Introduction to different logic families, Structure and operations of TTL, MOS and CMOS logic families

UNIT-IV

Introduction to Sequential Logic Circuits : Basic sequential circuits: SR-latch,D-latch,D flip- flop,JK flip- flop, T flip-flop, Conversion of basic flip-flops

UNIT-V

Sequential Logic Circuits Applications : Registers: Operation of all basic Shift Registers, Counters: Design of Asynchronous and Synchronous counters Ring counter and Johnson ring counter

UNIT-VI

Memory : Read-only memory, read/write memory - SRAM and DRAM, PLAs and their applications, Sequential PLDs and their applications, Introduction to field programmable gate arrays, PALs and their applications

Converters : Analog to Digital Converter, Digital to Analog Converter

RECOMMENDED READINGS:

1. DIGITAL DESIGN PRINCIPLES AND PRACTICES BY JOHN F. WAKERLY, PEARSON
2. DIGITAL FUNDAMENTALS BY THOMAS L. FLOYD , R. P JAIN, PEARSON
3. DIGITAL LOGIC DESIGN BY MORRIS MANO, PEARSON
4. DIGITAL ELECTRONICS LOGIC AND DESIGN BY CHERRY BHARGAVA, BS PUBLICATIONS

ECE216: DIGITAL ELECTRONICS LABORATORY

L: 0 T: 0 P: 2 Credits: 1

COURSE OUTCOMES: Through this course students should be able to

CO1 :: describe the design and functionality of digital circuits.

CO2 :: analyze the digital circuits and compare its theoretical performance to actual performance.

CO3 :: analyze functionality of the digital trainer kit to verify basic logic truth table.

LABORATORY WORK:

Analysis and Synthesis of Boolean Expressions using Basic Logic Gates

- Understanding the combinational logic by implementing the Boolean function using basic logic gates

Analysis and Synthesis of Arithmetic Expressions using Adders/Subtractors

- To design and analyze the circuit for Full adder and Full subtractor using Logic Gates.
- Analysis and Synthesis of Logic Functions using Multiplexers and decoders
- Understanding the combinational logic by implementing the Boolean function using multiplexer
- Understanding the combinational logic by implementing the Boolean function using Decoder

Analysis and Synthesis of Sequential Circuits using Flip-Flops

- Understanding the sequential logic by implementing the flip flop with the help of logic gates
- Understanding the sequential logic by implementing the counter with flip flop.
- Analysis of Functions of BCD-TO-7-segment Decoder / Driver and Operation of 7-segment LED Display
- To visualize the output of decade counter on seven segment display

Design and implementation of combinational and sequential circuit using Software

- To implement and simulate combinational and sequential circuit using DSCH/Proteus.
- Design and Implementation of application based projects, any two to be implemented
- To design 4 bit digital calculator which can perform addition and multiplication and display using 7 segment.
- To design a circuit which can generate random number and display using 7 segment.
- To design a circuit for smart home automation.
- To design a circuit for secure locking mechanism.
- To design a circuit for global positioning system synchronize clock.
- To design a system for solar tracking.
- To design a up and down fading lights (different colored LEDs) with specified delays using flip flops/counters
- Design a universal counter which can perform different shift operations using multiplexer.
- Design a digital calculator which can implement subtraction and division functions, and display output in 7-segment display unit

RECOMMENDED READINGS:

1. DIGITAL DESIGN PRINCIPLES AND PRACTICES BY JOHN F. WAKERLY, PEARSON
2. DIGITAL FUNDAMENTALS BY THOMAS L. FLOYD , R. P JAIN, PEARSON
3. DIGITAL ELECTRONICS PRINCIPLES AND INTEGRATED CIRCUITS BY ANIL K MAINI, PEARSON

MEC107: BASIC ENGINEERING MECHANICS

L: 3 T: 1 P: 0 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: discuss and Understand and analyze forces, moments and their applications in real life situations.

CO2 :: describe and Understand the basic concepts of the laws of friction and moment of inertia and its real life applications.

CO3 :: analyze and determine the forces in members of trusses and frames.

CO4 :: apply fundamental concepts of kinematics and kinetics of particles and rigid bodies to the analysis of practical problems

UNIT-I

Introduction to Mechanics: Basic concepts, System of forces, Coplanar Concurrent Forces, Components in 2-D Plane-Resultant-Moment of Forces and its Applications, Couples and Resultant of Force System, Equilibrium of System of forces, Free body diagrams, Equations of Equilibrium of Co- planar Systems

UNIT-II

Friction: Introduction to friction, Types of friction, Limiting friction, Angle of friction, Laws of Friction, Static and Dynamic friction, Motion of bodies

UNIT-III

Centroid and Moment of Inertia: Centroids of areas and lines, Centroids of composite plates and wires, First moments of areas and lines, Moment of inertia of plane sections, Theorems of moment of inertia, Center of gravity, Moment of inertia of standard and composite sections, Mass moment of inertia of thin plates

UNIT-IV

Analysis of structures: Introduction to trusses, Definition of trusses, Simple trusses, Analysis of truss by method of joint, Analysis of truss by method of section.

UNIT-V

Introduction to Dynamics: Basic terms, general principles in dynamics, Types of motion, General Plane motion, Rectilinear motion, Plane curvilinear motion

UNIT-VI

Plane Kinematics and Kinetics of Rigid bodies: D' Alembert's principle and its applications in plane motion and connected bodies, Work energy principle and its application in plane motion of connected bodies, Kinetics of rigid body rotation

RECOMMENDED READINGS:

1. VECTOR MECHANICS FOR ENGINEERS, STATICS AND DYNAMICS BY BEER AND JOHNSTON, MCGRAW HILL EDUCATION.
2. ENGINEERING MECHANICS: STATICS BY ANDREW PYTEL | JAAN KIUSALAAS, CENGAGE LEARNING.
3. ENGINEERING MECHANICS BY BASUDDEB BHATTACHARYYA, OXFORD UNIVERSITY PRESS.
4. ENGINEERING MECHANICS - STATICS AND DYNAMICS BY R. C. HIBBELER, A. GUPTA, PEARSON.
5. ENGINEERING MECHANICS - STATICS AND DYNAMICS BY S K SINHA, PEARSON.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

MTH166: DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

L: 3 T: 1 P: 0 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: define and distinguished between different types of differential equations.

CO2 :: understand the use of different methods for the solution of differential equations.

CO3 :: apply the important concepts associated with derivatives of vector fields such as Gradient, divergence, curl, and scalar potential etc.

CO4 :: analyze the second-order partial differential equations such as the heat, wave and Laplace equation.

CO5 :: evaluate the line, surface, volume integral using various theorems of vector calculus.

UNIT-I

Ordinary differential equations: exact equations, equations reducible to exact equations, equations of the first order and higher degree, Clairaut's equation.

UNIT-II

Differential equations of higher order : introduction to linear differential equation, Solution of linear differential equation, linear dependence and linear independence of solution, method of solution of linear differential equation- Differential operator, solution of second order homogeneous linear differential equation with constant coefficient, solution of higher order homogeneous linear differential equations with constant coefficient.

UNIT-III

Linear differential equation : solution of non-homogeneous linear differential equations with constant coefficients using operator method, method of variation of parameters, method of undetermined coefficient, solution of Euler-Cauchy equation, simultaneous differential equations by operator method.

UNIT-IV

Partial differential equation : introduction to partial differential equation, method of Separation of Variables, solution of wave equation, solution of heat equation, solution of Laplace equation.

UNIT-V

Vector calculus I: Limit continuity and differentiability of vector functions, length of space curve, motion of a body or particle on a curve, gradient of a scalar field and directional derivatives, divergence and curl of vector field.

UNIT-VI

Vector calculus II: line integral, Greens' theorem, surface area and Surface integral, Stokes' theorem, Gauss's divergence theorem.

RECOMMENDED READINGS:

1. ADVANCED ENGINEERING MATHEMATICS BY R.K.JAIN, S.R.K. IYENGER AND NAROSA PUBLISHING HOUSE.
2. HIGHER ENGINEERING MATHEMATICS BY DR. B.S. GREWAL, KHANNA PUBLISHERS.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

PEL121: COMMUNICATION SKILLS-I

L: 1 T: 2 P: 1 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: analyze their grammatical and communicative competence

CO2 :: associate the grammatical components in written and verbal communication

CO3 :: employ their vocabulary to choose language that precisely expresses their meanings

CO4 :: develop self-confidence with improved command over the English language

CO5 :: apply correct punctuation in writing to derive meaningful sentences

CO6 :: construct appropriate sentences in speech and writing using tense according to context

UNIT-I

Parts of speech: introduction to the basics of parts of speech

UNIT-II

Articles, determiners, and quantifiers: rules to use and omit articles, usage of determiners and quantifiers

UNIT-III

Tenses and conditionals: introduction to tenses and usage of tenses, introduction to conditionals and its usage

UNIT-IV

Sentence and clauses: introduction to different types of sentences and clauses

UNIT-V

Modals and punctuation: modals and their usage, usage of punctuation marks

UNIT-VI

Vocabulary: phrasal verbs, confusing words and idioms

RECOMMENDED READINGS:

1. ENGLISH GRAMMAR IN USE BY RAYMOND MURPHY, CAMBRIDGE UNIVERSITY PRESS
2. INTERMEDIATE ENGLISH GRAMMAR BY RAYMOND MURPHY, CAMBRIDGE UNIVERSITY PRESS

CSE205: DATA STRUCTURES AND ALGORITHMS

L: 3 T: 0 P: 2 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: design, implement and study various data structure algorithms.

CO2 :: understand the importance of data structures in context of writing efficient programs.

CO3 :: develop skills to apply appropriate data structures in problem solving.

CO4 :: formulate new solutions for programming problems or improve existing code using learned algorithms and data structures

UNIT-I

Introduction: Basic Data Structures, Basic Concepts and Notations, Complexity analysis: time space and trade off, Omega Notation, Theta Notation, Big O notation

Arrays: Linear arrays: memory representation, Traversal, Insertion, Deletion, Searching, Merging and their complexity analysis.

Sorting and Searching: Bubble sort, Insertion sort, Selection sort

UNIT-II

Linked Lists: Introduction, Memory representation, Allocation, Traversal, Insertion, Deletion, Header linked lists: Grounded and Circular, Two-way lists: operations on two way linked lists

UNIT-III

Stacks: Introduction: List and Array representations, Operations on stack (traversal, push and pop), Arithmetic expressions: polish notation, evaluation and transformation of expressions. Evaluation and transformation of expressions, Function call

Queue: Array and list representation, operations (traversal, insertion and deletion), Priority Queues, Deques

UNIT-IV

Recursion: Introduction, Recursive implementation of Towers of Hanoi, Merge sort, Quick sort

Trees : Binary trees - introduction (complete and extended binary trees), memory representation (linked, sequential), Pre-order traversal, In-order traversal, Post-order traversal using recursion, Binary Search Tree- searching, insertion, deletion

UNIT-V

AVL trees and Heaps: AVL trees - introduction, AVL trees Insertion, AVL trees Deletion, Heaps - Insertion, Heapify, Deletion, Heap Sort, Huffman algorithm

UNIT-VI

Graphs: Warshall's algorithm, Shortest path algorithm Floyd Warshall Algorithm (modified Warshall algorithm), Graph Traversal: BFS, DFS

Hashing : Hashing Introduction, Hash Functions, Hash Table, Closed hashing (open addressing), Linear Probing, Quadratic Probing, Double Hashing, Open hashing (separate chaining)

Assistant Dean
Division of Academic Affairs
Lovely Professional University

LABORATORY WORK:

- **Arrays:** Program to implement insertion and deletion operations in arrays
- **Searching:** Program to implement different searching techniques - linear and binary search
- **Sorting:** Program to implement different sorting techniques–bubble, selection and insertion sort
- **Linked List:** Program to implement searching, insertion and deletion operations in linked list
- **Doubly Linked List:** Program to implement searching, insertion and deletion operations in doubly linked list
- **Stacks:** Program to implement push and pop operations in stacks using both arrays and linked list
- **Queues:** Program to implement enqueue and dequeue operations in queues using both arrays and linked list
- **Recursions:** Program to demonstrate concept of recursions with problem of tower of Hanoi
- **Recursive Sorting:** Program to implement recursive sorting techniques - merge sort, quick sort
- **Trees:** Program to create and traverse a binary tree recursively
- **Binary Search Tree:** Program to implement insertion and deletion operations in BST
- **Heaps:** Program to implement insertion and deletion operations in Heaps and Heap Sort

RECOMMENDED READINGS:

1. DATA STRUCTURES BY SEYMOUR LIPSCHUTZ, MCGRAW HILL EDUCATION
2. DATA STRUCTURES AND ALGORITHMS BY ALFRED V. AHO, JEFFREY D. ULLMAN AND JOHN E. HOPCROFT, PEARSON

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE211: COMPUTER ORGANIZATION AND DESIGN

L: 3 T: 1 P: 0 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: review the structure and functioning of a digital computer and understand its overall system architecture.

CO2 :: describe and understand the generic principles that underlie the building of a digital computer, digital logic and memory hierarchy

CO3 :: analyze the working of memory unit and study the examples of mapping techniques for different cache memory systems

CO4 :: understand functioning of the basic building blocks of a computer

CO5 :: visualize the underlying architecture and connection of various hardware components of a computer

CO6 :: develop innovative architectural designs of computers based on the common and fundamental concepts

UNIT-I

Basics Of Digital Electronics: Multiplexers and De multiplexers, Decoder and Encoder, Registers., Logic gates, Flip flops, binary counters, Introduction to combinational circuit, introduction to sequential circuits

Register Transfer and Micro Operations: Register Transfer Language and Register Transfer, Logic Micro Operations Shift Micro Operations register transfer arithmetic micro operations

UNIT-II

Computer Organization: instruction codes, computer registers, common bus system, computer instructions, timing and control, instruction cycle, memory reference instructions, input-output and interrupt

UNIT-III

Central Processing Unit: General Register Organization, Data Transfer and Manipulation, Program control, Addressing Modes, Reduced instruction set computer, Complex instruction set computer

UNIT-IV

Input-Output Organization: Input Output Interface, Priority interrupt, Data transfer schemes, Direct memory access transfer Input/output processor. modes of data transfer

UNIT-V

Memory hierarchy: main memory, auxiliary memory, associative memory, cache memory, virtual memory

UNIT-VI

Introduction to Parallel Processing: Pipelining, Characteristics of multiprocessors, Interconnection Structures parallel processing

Latest technology and trends in computer architecture: multi-cores processor., next generation processors architecture microarchitecture latest processor for smartphone or tablet and desktop

Multiprocessors: Categorization of multiprocessors (SISD,MIMD,SIMD.SPMD), Introduction to GPU

Assistant Dean
Division of Academic Affairs
Lovely Professional University

RECOMMENDED READINGS:

1. COMPUTER SYSTEM ARCHITECTURE BY MORRIS MANO, PRENTICE HALL
2. COMPUTER ARCHITECTURE A QUANTITATIVE APPROACH BY HENNESSY,J.L,DAVID A PATTERSON, AND GOLDBERG, PEARSON
3. COMPUTER ORGANIZATION AND ARCHITECTURE-DESIGNING FOR PERFORMANCE BY WILLIAM STALLINGS, PRENTICE HALL

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE320: SOFTWARE ENGINEERING

L: 3 T: 0 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: plan and deliver an effective software engineering process, based on knowledge of widely used development life cycle models.

CO2 :: construct implementable design from requirement specification, following a structured and organized process.

CO3 :: translate a requirements specification into an implementable design, following a structured and organized process.

CO4 :: formulate a testing strategy for a software system, employing test case design techniques such as functional and structural testing.

CO5 :: analyze project including planning, scheduling, estimation and configuration management.

UNIT-I

Introduction to software engineering : Evolution and impact of software engineering, Software life cycle models, Waterfall model, Prototyping model, Evolution and spiral models, Feasibility study, Functional and non-functional requirements, Requirement gathering, Requirement analysis and specification

UNIT-II

Issues in software design : Basic issues in software design, Modularity, Cohesion, Coupling and layering Function oriented software design Data flow diagram and structure chart

UNIT-III

Object modelling : User interface design, unified process, Object modelling using UML, use case model development, Coding standards and code review techniques

UNIT-IV

Testing : Fundamentals of testing, Black box testing techniques, White box testing techniques, Levels of testing Test cases

Introduction to selenium : Feature of selenium, Versions of selenium, Record and play back

UNIT-V

Software project management : Project management, Project planning and control, Cost estimation Project scheduling using PERT and GANTT charts Software configuration management

UNIT-VI

Quality management : Quality management, ISO and SEI CMMI, PSP and six sigma, Computer aided software engineering, Software maintenance, Software reuse, Component based software development

Advance techniques of software engineering : Agile development methodology, Scrum, Aspect oriented programming, Extreme Programming, Adaptive software development, Rapid application development (RAD) Software cloning

Assistant Dean
Division of Academic Affairs
Lovely Professional University

RECOMMENDED READINGS:

1. FUNDAMENTALS OF SOFTWARE ENGINEERING BY RAJIB MALL, PRENTICE HALL
2. SOFTWARE ENGINEERING BY IAN SOMMERVILLE, PEARSON
3. SOFTWARE ENGINEERING:A PRACTITIONER APPROACH BY ROGER S.PRESSMAN, MCGRAW HILL EDUCATION
4. SOFTWARE ENGINEERING FUNDAMENTALS BY ALI BEHFOROOZ AND FREDERICKS J. HUDSON OXFORD UNIVERSITY PRESS

Assistant Dean
Division of Academic Affairs
Lovely Professional University

INT213: PYTHON PROGRAMMING

L: 2 T: 0 P: 2 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: analyze real life situational problems and think creatively about solutions of them

CO2 :: determine the methods to create and manipulate Python programs by utilizing the data structures like lists dictionaries tuples and sets

CO3 :: articulate the Object-Oriented Programming concepts such as encapsulation, inheritance and polymorphism as used in Python

CO4 :: develop the ability to write database applications in Python

CO5 :: analyze and visualize the data using python libraries

UNIT-I

Introduction: python programming language, introduction to program and debugging, formal and natural language

Variables, Expression and Statements: Values and types, variables, variables name and keywords, statements, operators and operand, order of operations, operations on string, composition and comments

Conditionals and Iteration: modulus operator, boolean expressions, logic operators, conditional, alternative execution, nested conditionals and return statements, while statements, encapsulation and generalization

Functions and recursion: function calls, type conversion and coercion, math functions, adding new function parameters and argument recursion and its use

UNIT-II

String: string a compound data type, length, string traversal, string slices, comparison, find function, looping and counting

Lists: list values, length, membership, operations, slices, deletion, accessing elements, list and for loops list parameters and nested list

Tuples and Dictionaries: mutability and tuples, tuple assignment, tuple as return values, random numbers and list of random numbers, counting and many buckets, dictionaries operations and methods sparse matrices aliasing and coping

UNIT-III

Files and exceptions: text files, writing variables, directories, pickling, exceptions, glossary

Building GUI using python: tkinter programming, tkinter widgets like button, canvas, entry, frame, label, list box, menu, message, scale, text, spinbox, labelframe, tkMessagebox, standard attributes, geometry management, GUI and database with sqlite3

UNIT-IV

Classes and objects: creating classes, creating instance objects, accessing attributes, overview of OOP terminology

Object oriented programming terminology: Class Inheritance, Overriding Methods, Data Hiding, Function Overloading

Assistant Dean
Division of Academic Affairs
Lovely Professional University

UNIT-V

Data visualization with matplotlib: line plot, multiple subplots in one figure, histograms, bar charts, pie charts, scatter plots

Handling data with pandas: series, dataframes, read and write csv file, operations using dataframe

UNIT-VI

Advanced machine learning libraries: numpy - datatype, array operations, statistical functions, broadcasting, scipy - cluster, interpolate, ndimage, application and usage of tensor flow, keras, scikit learn libraries

LABORATORY WORK:

- Program 1: Program to enter two numbers and print the arithmetic operations like +,-,*, /, // and %.
- Program 2: Write a program to find whether an inputted number is perfect or not. Program 3: Write a Program to check if the entered number is Armstrong or not. Program 4: Write a Program to find factorial of the entered number.
- Program 5: Write a Program to enter the number of terms and to print the Fibonacci Series. Program 6: Write a Program to enter the string and to check if it's palindrome or not using loop. Program 7: Recursively find the factorial of a natural number.
- Program 8: Read a file line by line and print it.
- Program 9: Remove all the lines that contain the character "a" in a file and write it into another file.
- Program 10 Read a text files and display the number of vowels/consonants/uppercase/lowercase characters in the file.
- Program 11 Create a binary file with name and roll no. Search for a given roll number and display the name, if not found display appropriate message.
- Program 12 Write a random number generator that generates random numbers between 1 and 6(simulates a dice).
- Program 13 Write a python program to implement a stack using a list data structure. Program 14 Take a sample of ten phishing e-mails (or any text file) and find most common Program 15 Read a text file line by line and display each word separated by a #
- Program 16 Create a student table and insert data. Implement the following SQL commands on the student table:
- ALTER table to add new attributes / modify data type / drop attribute.
- UPDATE table to modify data.
- ORDER By to display data in ascending / descending order.
- DELETE to remove tuple(s).
- GROUP BY and find the min, max, sum, count and average Program 17: Integrate SQL with Python by importing the MySQL module.

RECOMMENDED READINGS:

1. INTRODUCTION TO PROGRAMMING USING PYTHON BY Y. DANIEL LIANG, PEARSON.
2. PYTHON PROGRAMMING: USING PROBLEM SOLVING APPROACH BY REEMA THAREJA, OXFORD UNIVERSITY PRESSES.

INT306: DATABASE MANAGEMENT SYSTEMS

L: 0 T: 0 P: 5 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: develop skills and understanding in the database design and make use of database management systems for applications.

CO2 :: develop understanding about relational algebra, relational model and SQL for implementing and maintaining databases.

CO3 :: develop understanding about the different issues involved in the design and implementation of a database system.

CO4 :: develop skills and understanding about the real time transaction management systems and the concurrency control techniques.

CO5 :: compose programming constructs such as functions, stored procedures and triggers that can be shared by multiple forms, reports and data management applications.

LABORATORY WORK:

Introduction to Databases

- purpose of database systems, components of dbms, applications of dbms
- three tier dbms architecture
- data independence, database schema, instance
- data modeling, entity relationship model, relational model

Relational Query Languages

- relational algebra
- introduction to data definition language, data manipulation
- data control and transaction control language
- integrity constraints
- database keys
- sql basic operations
- aggregate functions
- sql joins
- set operators, views
- sub queries

Relational Database Design

- data integrity rules, functional dependency
- need of normalization, first normal form, second normal form
- third normal form, boyce codd normal form
- multivalued dependencies, fourth normal form
- join dependencies, fifth normal form and pitfalls in relational database design

Database Transaction Processing

- transaction system concepts, desirable properties of transactions
- schedules, serializability of schedules
- concurrency control
- recoverability

Assistant Dean
Division of Academic Affairs
Lovely Professional University

Programming Constructs in Databases

- flow control statements
- functions, stored procedures
- cursors
- triggers
- exception handling

File Organization and Trends in Databases

- file organizations and its types
- indexing, types of indexing
- hashing, hashing techniques
- introduction to big data, nosql systems

RECOMMENDED READINGS:

1. DATABASE SYSTEM CONCEPTS BY HENRY F. KORTH, ABRAHAM SILBERSCHATZ, S.SUDARSHAN, MCGRAW HILL EDUCATION.
2. DATABASE SYSTEMS: MODELS, LANGUAGES, DESIGN AND APPLICATION PROGRAMMING BY RAMEZ ELMASRI, SHAMKANT B. NAVATHE, PEARSON.
3. AN INTRODUCTION TO DATABASE SYSTEMS BY C. J. DATE, S. SWAMYNATHAN, A. KANNAN, PEARSON.
4. SQL, PL/SQL: THE PROGRAMMING LANGUAGE OF ORACLE BY IVAN BAYROSS, BPB PUBLICATIONS.
5. SIMPLIFIED APPROACH TO DBMS BY PRATEEK BHATIA AND GURVINDER SINGH, KALYANI PUBLISHERS.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

MGN231: PROJECT

L: 0 T: 0 P: 4 Credits: 2

COURSE OUTCOMES: Through this course students should be able to

CO1 :: identify the issues related to community development and to find solutions for improvement

CO2 :: develop creative thinking process and evolving behavioral changes for effective solution to identified problem

CO3 :: employ the skills to promote unique idea among public aiming to originate substantial and sustainable development bringing in difference in the community at large

CO4 :: illustrate the ability to deliver effective presentation and written report covering major issues and suggestions related to the Project

Description:

The purpose is to provide an opportunity as a trainee to seek, identify and further develop the appropriate level of professionalism while observing the professional dealings at respective industry/organization and learn the latest techniques and skills to build a strong foundation for their career growth.

Course Weightages:

ATT	CA	MTP	ETP1	ETP2
0	0	0	100	0

Attendance Requirement: NA

Continuous Assessment (CA): NA

ETP Evaluation Parameters:

Parameter	Marks
Action Taken to achieve Objectives and their supportive documents including Photographs before and after, videos and testimonials	20
Cause of the Problem Identified	10
Effectiveness of the Project	10
Innovativeness and Uniqueness of the Project undertaken	20
Objective	10
Presentation	20
Problem identification	10
Total	**100**

Applicability of R Grade: Yes

The conduct of the course shall be as per guidelines issued by the University for the given course.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

MTH401: DISCRETE MATHEMATICS

L: 3 T: 0 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: understand various ways to prove or disapprove some logical statements.

CO2 :: solve recurrence relation by using different methods.

CO3 :: describe the concept of graphs and their properties.

CO4 :: understand various concepts of number theory and its applications.

UNIT-I

Logic and Proofs: Propositional logic, propositional equivalences, quantifiers, Introduction to proof, direct proof, proof by contraposition, vacuous and trivial proof, proof strategy, proof by contradiction, proof of equivalence and counterexamples, mistakes in proof.

UNIT-II

Recurrence relations: recurrence relation, modelling with recurrence relations, homogeneous linear recurrence relations with constant coefficients, Method of inverse operator to solve the non- homogeneous recurrence relation with constant coefficient, generating functions, solution of recurrence relation using generating functions.

UNIT-III

Counting principles and relations: principle of Inclusion-Exclusion, Pigeonhole, generalized pigeonhole principle, relations and their properties, combining relation, composition, representing relation using matrices and graph, equivalence relations, partial and total ordering relations, lattice, sub lattice, Hasse diagram and its components.

UNIT-IV

Graphs theory I: graph terminologies, special types of graphs(complete, cycle, regular, wheel, cube, bipartite and complete bipartite), representing graphs, adjacency and incidence matrix, graph- isomorphism, path and connectivity for undirected and digraphs, Dijkstra's algorithm for shortest path problem.

UNIT-V

Graphs theory II: planner graphs, Euler formula, colouring of a graph and chromatic number, tree graph and its properties, rooted tree, spanning and minimum spanning tree, decision tree, infix, prefix, and postfix notation.

UNIT-VI

Number theory and its application in cryptography: divisibility and modular arithmetic, primes, greatest common divisors and least common multiples, Euclidean algorithm, Bezout's lemma, linear congruence, inverse of (a modulo m), Chinese remainder theorem, encryption and decryption by Ceasar cipher and affine transformation, Fermat's little theorem.

RECOMMENDED READINGS:

1. DISCRETE MATHEMATICS AND ITS APPLICATIONS BY KENNETH H ROSEN, MCGRAW HILL EDUCATION.
2. HIGHER ENGINEERING MATHEMATICS BY B. V. RAMANA, MC GRAW HILL.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

PEL131:COMMUNICATION SKILLS-II

L: 3 T: 1 P: 0 Credits: 4

Course Outcomes: Through this course students should be able to

CO1 :: compose simple connected text on topics which describes experiences and impression in written discourse.

CO2 :: understand the main points of standard speech on familiar matters regularly encountered at work.

CO3 :: express unprepared speech into conversation on topics that are familiar, of personal interest or pertinent to everyday life.

CO4 :: describe experiences and events, dreams, hopes and ambitions by composing phrases in a simple way.

UNIT-I

Meeting and greeting people : vocabulary and common errors related to salutation, vocabularyand common errors related to self-introduction, vocabulary and common errors related to asking for help, common errors related to tenses and parts of speech

UNIT-II

Usage of connectors and transition words in conversation:usage of connectors, transitionwords and vocabulary related to routine, usage of connectors,transition words and vocabulary related to shopping, usage of connectors,transition words and vocabulary related to vacation

UNIT-III

Engaging in small talk : direct and indirect speech, vocabulary and phrases related to small talk, importance of small talk

UNIT-IV

Presenting your ideas effectively : introducing stress and intonation, introducing dignitaries using positive adjectives, presenting ideas on products using positive adjectives

UNIT-V

Paragraph writing and power point presentation : introducing paragraph writing, key elements of paragraph writing, usage of collocations, do's and don'ts of power point presentation

UNIT-VI

Making reservation and arrangements: telephone etiquettes, vocabulary and phrases for makingreservation and arrangements, formal letter writing- request and complaint letters

RECOMMENDED READINGS:

1. ENGLISH GRAMMAR IN USE BY RAYMOND MURPHY, CAMBRIDGE UNIVERSITY PRESS

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE306: COMPUTER NETWORKS

L: 3 T: 0 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: describe the importance of data communications and the Internet in supporting business communications and daily activities

CO2 :: analyze the different types of network devices and their functions within a network

CO3 :: describe the basic protocols of computer networks, and how they can be used to assist in network design and implementation

CO4 :: show the practical utilization of networking standards and protocols in relevant scenarios

UNIT-I

INTRODUCTION : Networks and Types, Uses of Computer Networks, Network software architecture and its layers and protocols, Network hardware architecture and its topologies and device like HUB, Switch and Routers

NETWORK MODELS : Protocol Layering, OSI Model, TCP/IP protocol suite

UNIT-II

PHYSICAL LAYER: Signal & Media : Basics for Data Communications and Analog and Digital signals, Transmission Impairments and Performance, Data Rate, Transmission media like Guided and Unguided media Cabling standards

PHYSICAL LAYER: Modulation & Multiplexing : Digital to Digital Conversion, Analog to Digital Conversion Analog to Analog conversion Digital to Analog conversion Multiplexing

UNIT-III

DATA LINK LAYER : Data link Layer design issues, Elementary Data link Protocols, Error Detection and Correction- Hamming code CRC Parity Checksum Switch working

MAC SUBLAYER : Multiple Access Protocols- ALOHA, CSMA and CSMA/CD, Random Access, Controlled access Ethernet protocol

UNIT-IV

NETWORK LAYER: IP Addressing : Network layer design issue, IP Addressing Both Class full and Classless, Sub netting and Super netting, Sub netting examples, Network layer services, Network layer performance Forwarding of IP packets IP Header IPv6 addressing

UNIT-V

NETWORK LAYER: Routing : Routing Algorithm-Shortest path algorithm, Distance vector Routing, Link State routing, Routing algorithms, Unicast routing protocols

NETWORK LAYER: Congestion Control : Congestion Control Algorithms

UNIT-VI

TRANSPORT LAYER : Transport Layer Services, TCP- Header format and handshaking operation, UDP- Header format

APPLICATION LAYER : Domain Name System, E Mail, FTP

RECOMMENDED READINGS:

1. DATA COMMUNICATIONS AND NETWORKING BY BEHROUZ FOROUZAN, MCGRAW HILL EDUCATION
2. COMPUTER NETWORKS BY ANDREW S. TANENBAUM, PEARSON

CSE307: INTERNETWORKING ESSENTIALS

L: 0 T: 0 P: 2 Credits: 1

COURSE OUTCOMES: Through this course students should be able to

CO1 :: relate the hardware and components of a network and the interrelations

CO2 :: interpret network layer routing protocols

CO3 :: examine concepts and theories of networking to apply them to various situations and classifying networks

CO4 :: show an end to end connectivity using network commands

LABORATORY WORK:

Network hardware and IP addressing concept

- Working of hub, switch and Router, Adding of interfaces in devices
- Cabling - Creation of straight and Cross cable using crimping tool
- IP addressing basics, configuration using CLI, VLSM and FLSM on single router
- Implementation of Star, Mesh, Bus and Hybrid Topology

Network Commands

- Ping, tracert, arp, netstat, ipconfig, ftp, nslookup, snmpget, snmpgetbulk and snmpset (use DOS and scenario based configuration)

Network layer routing protocols

- Implementation of Static Routing using Class full and classless (FLSM)
- Implementation of Static Routing using VLSM
- Routing information Protocol(RIP) using class full and classless (FLSM)
- Routing information Protocol(RIP) using VLSM
- Server Configuration and LAN Setup
- Implementation of FTP, Implementation of HTTP and Email setup on server
- Implementation of DNS, Implementation of DHCP
- Implementation of LAN with configuration of inter-networking devices and any application layer protocol

IPv6 addressing and routing

- IPv6 Addressing & Stateless Address Auto Configuration (SLAAC)
- IPv6 Neighbor Discovery
- IPv6 Static Routing
- IPv6 Dynamic Routing

RECOMMENDED READINGS:

1. COMPUTER NETWORKS BY ANDREW S. TANENBAUM, PEARSON
2. DATA COMMUNICATION AND NETWORKING BY BEHROUZ A. FOROUZAN, MCGRAW HILL EDUCATION

CSE310: PROGRAMMING IN JAVA

L: 3 T: 0 P: 2 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: identify basic constructs of Java programming and applying them to solve the real-world

CO2 :: applying the Object-oriented programming principles to write efficient and reusable codes

CO3 :: understanding the application of abstract classes, interfaces and Lambda expressions

CO4 :: developing robust java applications to handle environment specific issues at run-time

CO5 :: using pre predefined java libraries and in-built data structures to develop efficient java applications

UNIT-I

Java Platform Overview: Defining how the Java language achieves platform independence, Differentiating between the Java ME, Java SE, and Java EE Platforms, Evaluating Java libraries, middle-ware, and database options, Defining how the Java language continues to evolve

What Is a Java Program ?: Introduction to Computer Programs, Key Features of the Java Language, The Java Technology and Development Environment, Running/testing a Java program

Creating a Java Main Class: Java Classes, The main Method

Data In the Cart: Introducing variables, Working with Strings, Working with numbers, Manipulating numeric data

Managing Multiple Items: Working with Conditions, Working with a List of Items, Processing a list of items

UNIT-II

Manipulating and Formatting the Data in Your Program: Using the String Class, Using the String Builder Class, More about primitive data types, The remaining numeric operators, Promoting and casting variables

More on Conditionals: Relational and conditional operators, More ways to use if/else constructs, Using Switch Statements

More on Arrays and Loops: Working with Dates, Parsing the args Array, Two-dimensional Arrays, Alternate Looping Constructs, Nesting Loops, The Array List class

Describing Objects and Classes: Working with objects and classes, Defining fields and methods, Declaring, Instantiating, and Initializing Objects, Working with Object References, Doing more with Arrays

Creating and Using Methods: Using methods, Method arguments and return values, Static methods and variables, How Arguments are Passed to a Method, Overloading a method

Assistant Dean
Division of Academic Affairs
Lovely Professional University

UNIT-III

Using Encapsulation: Access Control, Encapsulation, Overloading constructors

Using Inheritance: Overview of inheritance, Working with subclasses and super-classes, Overriding methods in the superclass, Introducing polymorphism, Creating and extending abstract classes, Modeling business problems using Java classes, Making classes immutable

Overriding Methods, Polymorphism, and Static Classes: Using access levels: private, protected, default, and public, Overriding methods, Using virtual method invocation, Using varargs to specify variable arguments, Using the instance of operator to compare object types, Using upward and downward casts, Modeling business problems using the static keyword, Implementing the singleton design pattern

UNIT-IV

Abstract and Nested Classes: Designing general-purpose base classes by using abstract classes, Constructing abstract Java classes and subclasses, Applying final keyword in Java, Distinguish between top-level and nested classes

Using Interfaces: Polymorphism in the JDK foundation classes, Using Interfaces, Using the List Interface, Introducing Lambda expressions

Interfaces and Lambda Expressions: Defining a Java interface, Choosing between interface inheritance and class inheritance, Extending an interface, Defaulting methods, Anonymous inner classes, Defining a Lambda Expression

UNIT-V

Exceptions and Assertions: Handling Exceptions: An overview, Propagation of exceptions, Catching and throwing exceptions, Handling multiple exceptions and errors, Defining the purpose of Java exceptions, Using the try and throw statements, Using the catch, multi-catch, and finally clauses, Auto close resources with a try-with-resources statement, Recognizing common exception classes and categories, Creating custom exceptions, Testing invariants by using assertions

UNIT-VI

Collections and Generics: Creating a custom generic class, Using the type inference diamond to create an object, Creating a collection by using generics, Implementing an ArrayList, Implementing a TreeSet, Implementing a HashMap, Implementing a Deque, Ordering collections

LABORATORY WORK:

- **Creating a Java Main Class:** Program to implement a java class
- **Managing Multiple Items:** Program to demonstrate the use of list of items
- **Manipulating and Formatting the Data in Your Program:** Program to demonstrate the uses of String and String Builder
- **Describing Objects and Classes:** Program to demonstrate the instantiation of class and accessing the attributes using object of class
- **Using Inheritance:** Program to demonstrate the inheritance and its importance
- **Overriding Methods, Polymorphism, and Static Classes:** Program to implement polymorphism and using proper access control

- **Abstract and Nested Classes:** Program to demonstrate the use of abstract class and nested class
- **Interfaces and Lambda Expressions:** Program to demonstrate the inheritance through interfaces and use of Lambda Expressions
- **Eceptions and Assertions:**Program to demonstrate the use of all the keywords used for exception handling and need of assertion
- **I/O Fundamentals:** Program to implement read and write operation using console and **File Generics:** Program to define generic class and creating generic collection **Collections:** Program to implement Array List, Hash Map, Tree Set and Deque

RECOMMENDED READINGS:

1. INTRODUCTION TO JAVA PROGRAMMING BY Y. DANIEL LIANG, PEARSON
2. JAVA THE COMPLETE REFERENCE BY HERBERT SCHILDT, MCGRAW HILL EDUCATION

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE316: OPERATING SYSTEMS

L: 3 T: 0 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: recognize the basic structure of operating systems and classify roles and responsibilities of an operating System.

CO2 :: recognize the need and importance of fundamental concepts and principles of operating systems.

CO3 :: design the internal modules of an Operating System like memory management, process management disk management and inter process communication etc

UNIT-I

Introduction to Operating System : Operating System Operations and Functions, Multiprogramming and Multiprocessing System

Operating System Structure : System Calls

Process Management : Process states, Process scheduling, Operations on processes, Process concept, Life cycle, Process control box

UNIT-II

CPU Scheduling : CPU scheduler and dispatcher, Scheduling criteria, CPU scheduler - preemptive and non preemptive, Scheduling algorithms - process management in UNIX, First come first serve, Shortest job first, Round robin, Priority, Multi level feedback queue, multiprocessor scheduling, real time scheduling

UNIT-III

Threads : Overview, Multithreading Models

Process Synchronization : Critical Section Problem, Dining Philosopher Problem, Reader-writer Problem etc, Semaphores, Monitors, Synchronization hardware, Critical section problem - Two process solution Peterson's Solution

UNIT-IV

Deadlock : Deadlock Characterization, Handling, Handling of deadlocks- Deadlock Prevention, Deadlock Avoidance & Detection, Deadlock Recovery, Starvation, Critical regions

Information management : Files and directories, Directory structure, Directory implementation - linear list and hash table

File Management : Allocation methods, Free-Space Management

UNIT-V

Memory Management : Objectives and functions, Simple resident monitor program, Overlays - swapping, Schemes - Paging - simple and multi level, Fragmentation - internal and external, Virtual memory concept, Demand paging, Page interrupt fault, Page replacement algorithms, Segmentation - simple multi-level and with paging

UNIT-VI

Protection and Security : Need for Security, Different Security Environments, Application Security - Virus, Program Threats, Goals of protection, Principles of protection, Domain of protection, Access matrix System and network threats User authentication

Assistant Dean
Division of Academic Affairs
Lovely Professional University

Device management : Dedicated, shared and virtual devices, Serial access and direct access devices Disk scheduling methods Direct Access Storage Devices – Channels and Control Units

Inter process communication : Introduction to IPC (Inter process communication) Methods, Pipes open and close functions, Co-processes, Shared memory, Message queues, Passing File descriptors, Semaphores

RECOMMENDED READINGS:

1. OPERATING SYSTEM CONCEPTS BY ABRAHAM SILBERSCHATZ, GALVIN, WILEY
2. OPERATING SYSTEMS – INTERNALS AND DESIGN PRINCIPLES BY WILLIAMSTALLINGS, PRENTICE HALL

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE325: OPERATING SYSTEMS LABORATORY

L: 0 T: 0 P: 2 Credits: 1

COURSE OUTCOMES: Through this course students should be able to

CO1 :: examine and validate the various features of process and threads, including creation, replacing and termination

CO2 :: use of various Linux commands and system calls to understand how processes request operating system services

CO3 :: simulate the synchronization problems to ensure consistency of data using Mutex and Semaphores

CO4 :: examine inter-process communication using pipe, shared memory, and message passing mechanisms

LABORATORY WORK:

Introduction to Linux

- Basic Linux Commands: ls, cat, man, cd, touch, cp, mv, rmdir, mkdir, rm, chmod, pwd
- Simulation of Shell commands using system calls
- file/directory related system calls / library functions (read, write, open, close, lseek, opendir, readdir, closedir, etc)

Process creation and threading

- Creating processes
- Creating Threads
- Replacing process image using excel
- Process duplication using fork

Synchronization

- Synchronization with Mutexes
- Synchronization with semaphores
- Race Condition

Inter-process communication

- Pipes, popen and pclose functions
- Stream pipes, passing file descriptors
- Shared memory
- Message passing

RECOMMENDED READINGS:

1. BEGINING LINUX PROGRAMMING BY NEIL MATHEW & RICHARD STONES, WILEY
2. OPERATING SYSTEM CONCEPTS BY ABRAHAM SILBERSCHATZ, GALVIN, WILEY
3. ADVANCED PROGRAMMING IN THE UNIX ENVIRONMENT BY W.RICHARD STEVENS AND STEPHEN A. RAGO PEARSON
4. OPERATING SYSTEMS A DESIGN-ORIENTED APPROACH BY CHARLES CROWLEY, M.G.HILLS

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE408: DESIGN AND ANALYSIS OF ALGORITHMS

L: 3 T: 0 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: explain the basic techniques of analyzing the algorithms using space and time complexity, asymptotic notations

CO2 :: analyse various string matching algorithms and understand brute force algorithm design technique

CO3 :: understand divide and conquer algorithm design technique using various searching and sorting algorithms

CO4 :: define dynamic programming and greedy algorithm design technique and solve various all pair and single source shortest path problems

CO5 :: apply the backtracking method to solve some classic problems and understand branch and bound algorithm design technique

CO6 :: define various number theory problems and understand the basics concepts of complexity classes

UNIT-I

Foundations of Algorithm : Algorithms, Fundamentals of Algorithmic Problem Solving:, Basic Algorithm Design Techniques, Analyzing Algorithm, Fundamental Data Structure:, Linear Data Structure, Graphs and Trees, Fundamentals of the Analysis of Algorithm Efficiency:, Measuring of Input Size, Units for Measuring Running Time, Order of Growth, Worst-Case, Best-Case, and Average- Case Efficiencies, Asymptotic Notations and Basic Efficiency Classes:, O(Big-oh)-notation, Big-omega notation, Big-theta notation, Useful Property Involving the Asymptotic Notations, Using Limits for Comparing Orders of Growth

UNIT-II

String Matching Algorithms and Computational Geometry : Sequential Search and Brute-Force String Matching, Closest-Pair and Convex-Hull Problem, Exhaustive Search, Voronoi Diagrams, Naiva String-Matching Algorithm, Rabin-Karp Algorithm, Knuth-Morris-Pratt Algorithm

UNIT-III

Divide and Conquer and Order Statistics : Merge Sort and Quick Sort, Binary Search, Multiplication of Large Integers, Strassen's Matrix Multiplication, Substitution Method for Solving Recurrences, Recursion-Tree Method for Solving Recurrences, Master Method for Solving Recurrence, Closest-Pair and Convex-Hull Problems by Divide and Conquer, Decrease and Conquer: Insertion Sort, Depth-First Search and Breadth-First Search, Connected Components, Topological Sort, Transform and Conquer: Presorting, Balanced Search Trees, Minimum and Maximum, Counting Sort, Radix Sort, Bucket Sort, Heaps and Heap sort, Hashing, Selection Sort and Bubble Sort

UNIT-IV

Dynamic Programming and Greedy Techniques : Dynamic Programming: Computing a Binomial Coefficient, Warshall's and Floyd's Algorithm, Optimal Binary Search Trees, Knapsack Problem and Memory Functions, Matrix-Chain Multiplication, Longest Common Subsequence, Greedy Technique and Graph Algorithm: Minimum Spanning Trees, Prim's Algorithm, Kruskal's Algorithm, Dijkstra's Algorithm, Huffman Code, Single-source Shortest Paths, All-Pairs Shortest Paths, Iterative Improvement: The Maximum-Flow Problem Limitations of Algorithm Power: Lower-Bound Theory

UNIT-V

Backtracking and Approximation Algorithms : Backtracking: n-Queens Problem, Hamiltonian Circuit Problem, Subset-Sum Problem, Branch-and-Bound: Assignment Problem, Knapsack Problem, Traveling Salesman Problem Vertex-Cover Problem and Set-Covering Problem Bin Packing Problems

UNIT-VI

Number-Theoretic Algorithms and Complexity Classes : Number Theory Problems: Modular Arithmetic, Chinese Remainder Theorem, Greatest Common Divisor, Optimization Problems, Basic Concepts of Complexity Classes- P NP NP-hard NP-complete Problems

RECOMMENDED READINGS:

1. INTRODUCTION TO THE DESIGN AND ANALYSIS OF ALGORITHM BY ANANY LEVITIN, PEARSON
2. INTRODUCTION TO ALGORITHMS BY C.E. LEISERSON, R.L. RIVEST AND C. STEIN, THOMAS TELFORD LTD. THE DESIGN AND ANALYSIS OF COMPUTER ALGORITHMS BY A.V.AHO, J.E. HOPCROFT AND J.D.ULLMAN, PEARSON

INT404: ARTIFICIAL INTELLIGENCE

L: 3 T: 1 P: 0 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: define basic knowledge representation, problem solving, and learning methods of artificial intelligence.

CO2 :: understand approaches to syntax and semantics in NLP.

CO3 :: apply analytical concepts for solving logical problems using heuristics approaches.

CO4 :: develop AI learned concepts using python libraries.

UNIT-I

Introduction: What is intelligence?, History of AI, applications of AI, branches of AI, defining intelligence using turing test, making Machine think like human, building rational agent, general problem solver, building an intelligent agent, problem spaces and search, problem characteristics, production system and its characteristics

UNIT-II

Search techniques: search strategies, breadth first search, depth first search, generate and test, hill climbing, best first search, or graph, A* algorithm, constraint satisfaction, coloring problem, relation problem

UNIT-III

Knowledge representation and reasoning: knowledge based agents, knowledge representation techniques, propositional logic, predicate logic, clausal normal form, backward vs forward reasoning, introduction to logic programming understanding the building blocks of logic parsing a family tree

UNIT-IV

Natural Language Processing: introduction to NLP, phases of NLP, construction of parse tree, tokenizing text data, word stemming, word lemmatization, dividing text data into chunks, bag of words model, spell checking algorithm, soundex

UNIT-V

Building games with AI: using search algorithms in games, combinatorial search, minmax problem, alpha-beta pruning, negamax algorithm, last coin standing game, tic-tac-toe game, hexapawn game

UNIT-VI

Advanced topics in Artificial Intelligence: Types of Machine Learning, introduction to machine learning, activation functions, single neuron model, neural network architectures, genetic operators, genetic algorithm and its application, fuzzy logic, fuzzification and defuzzification, mamdani fuzzy inference system

RECOMMENDED READINGS:

1. ARTIFICIAL INTELLIGENCE BY KEVIN KNIGHT, ELAINE RICH, B. SHIVASHANKAR NAIR, MC GRAW HILL
2. ARTIFICIAL INTELLIGENCE: A MODERN APPROACH BY STUART RUSSEL, PETER NORVIG, PEARSON
3. ARTIFICIAL INTELLIGENCE WITH PYTHON BY PRATEEK JOSHI, PACKT PUBLISHING

MTH302: PROBABILITY AND STATISTICS

L: 3 T: 0 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: recall the basic principles of probability and Bayes theorem.

CO2 :: visualize and use the concept of random variables to find the probability of an event.

CO3 :: use of some important distributions to find the probabilities.

CO4 :: develop and test the hypothesis based on the nature of a problem.

UNIT-I

Basics of Probability: Probability of an Event, Rules of Probability, Conditional Probability and Independent Events, Bayes theorem.

UNIT-II

Random variables and its Characterization: Discrete and continuous random variables (in one dimension) and their distribution functions, Moments and Moment generating function of a random variable, Expectation and Variance of a random variable.

UNIT-III

Probability Distributions: Binomial Distribution, Poisson distribution, Moments and Moment generating function of Binomial Distribution, Moments and Moment generating function of Poisson distribution, Normal Distribution its Moments and M.G.F.

UNIT-IV

Point Estimation: Definition, Unbiased Estimators, Consistent Estimators, Sufficient Estimator, MLE (Method of Maximum Liklihood), Efficiency of estimations, Properties of Maximum likelihood.

UNIT-V

Hypothesis Testing: Types of Error, Goodness of a Fit, Student t-test, Chi- Square Test, Z-test, F- test.

UNIT-VI

Correlation and Regressions: Scatter plots, Coefficient of Correlation, Coefficient of Correlation for bi-variate data, Spearman's Rank Correlation Coefficient, Linear Regression and Properties of Regression Coefficients, Fitting of a curve.

RECOMMENDED READINGS:

1. FUNDAMENTALS OF MATHEMATICAL STATISTICS BY S.C.GUPTA AND V.K.KAPOOR, SULTAN CHAND & SONS (P) LTD.
2. PROBABILITY STATISTICS AND RANDAM PROCESSES BY T VEERARAJAN, MCGRAW HILL EDUCATION.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

PEV106: VERBAL ABILITY- I

L: 1 T: 2 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: analyze their grammatical and communicative competence

CO2 :: apply and use English grammar components appropriately in written communication

CO3 :: develop the ability to use the grammatical components in verbal communication

CO4 :: manage the varied language skills requirements of employers

UNIT-I

Subject verb agreement: subject, verb (a brief introduction), singular and plural nouns and verbs, rules of subject-verb agreement

UNIT-II

Vocabulary: introduction to root words, prefixes, and suffixes to understand words, commonly used words: used in newspapers, magazines, etc, common phrases used in corporate, antonyms/ synonyms

UNIT-III

Precise writing: idea elaboration, do's and do n'ts of precis writing

Sentence completion: type of questions- single and double blanks, eliminating options using verbal clues

UNIT-IV

Picture perception: picture perception and description

Para jumbles: types of Para jumbles, fixed and moving para jumbles, verbal and logical clues to solve Para jumbles

UNIT-V

Analogy: analogy questions, patterns of questions, common trick questions, eliminating options in analogy

UNIT-VI

Comprehension passages: techniques for smart reading - skimming, scanning and summarizing, types of questions, deducing author's tone and perspective

RECOMMENDED READINGS:

1. EFFECTIVE TECHNICAL COMMUNICATION BY M. ASHRAF RIZIVI, MC GRAW HILL

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CAP818: GAME TOOL-I

L:0 T: 0 P:3 Credits: 2

COURSE OUTCOMES: Through this course students should be able to

CO1 :: understand the relevance and infrastructure of the Multimedia Systems

CO2 :: memorize the multimedia concepts and various I/O Technologies

CO3 :: prepare frames and manage actions using the flash software

CO4 :: construct different animation and effects using flash software

LABORATORY WORK:

Flash and Its Objects

- Lab Work On Flash and Its Objects
- Creation Of Basic Animations Using Layering

Basic Animations

- Skewing
- Creation Of Basic Animations By Rotating
- How To Embed Flash Objects In HTML
- Scaling Features
- Lab Work On Skewing and Scaling Features

Animations Using Motion Presets

- Creation Of Basic Animations On Text and Graphics
- Lab Work On Flash
- Creation Of Basic Animations Using Twining Of Frames
- Creation Of Basic Animation Using Motion Presets
- Creation Of Basic Animations With Addition Of sound Files
- Lab Work On Flash and its Objects
- Motion Presets and Addition of Sound Files

Window Movie Maker

- Lab Work On Window Movie Maker For Creation Of Movies

Understanding Actions and Event Handlers

- Actions and Event Handlers
- Your First Five Actions
- Making Actions happen with event handlers

RECOMMENDED READINGS:

1. UNDERSTANDING MACROMEDIA FLASH 8 ACTIONSCRIPT 2: BASIC TECHNIQUES FORCREATIVES BY ALEX MICHAEL ANDREW RAPO MICHAEL RAPO, TAYLOR & FRANCIS
2. PPRINCIPLES OF MULTIMEDIA BY RANJAN PAREKH, MCGRAW HILL EDUCATION

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE322: FORMAL LANGUAGES AND AUTOMATION THEORY

L: 3 T: 0 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: analyze the fundamentals of theory of computation and design an infinite language in finite ways through deterministic finite automata, non-deterministic finite automata

CO2 :: apply an infinite language in finite ways through regular expressions and understanding the properties of regular languages

CO3 :: illustrate context free grammar and pushdown automata for a given Language

CO4 :: represent the acceptance of context free language using push down automata

CO5 :: develop a computational model using Turing machine for the given problem.

CO6 :: estimate the complexity for P and NP completeness for the given problem.

UNIT-I

FINITE AUTOMATA : Definition and Description of a Finite Automaton, Deterministic and Non- deterministic Finite State Machines, Transition Systems and Properties of Transition Functions, Acceptability of a String by a Finite Automaton, The Equivalence of DFA and NDFA, Mealy and Moore Machines, Minimization of Finite Automata, Basics of Strings and Alphabets, Transition Graph and Properties of Transition Functions, Regular Languages, The Equivalence of Deterministic and Non- deterministic Finite Automata

UNIT-II

REGULAR EXPRESSIONS AND REGULAR SETS : Regular Expressions and Identities for Regular Expressions, Finite Automata and Regular Expressions: Transition System Containing null moves, NDFA with null moves and Regular Expressions, Conversion of Non-deterministic Systems to Deterministic Systems, Algebraic Methods using Arden's Theorem, Construction of Finite Automata Equivalent to a Regular Expression, Equivalence of Two Finite Automata and Two Regular Expressions Closure Properties of Regular Sets, Pumping Lemma for Regular Sets and its Application, Equivalence between regular languages: Construction of Finite Automata Equivalent to a Regular Expression, Properties of Regular Languages, Non-deterministic Finite Automata with Null Moves and Regular Expressions Myhill-Nerode Theorem

UNIT-III

FORMAL LANGUAGES : Derivations and the Language Generated by a Grammar, Definition of a Grammar, Chomsky Classification of Languages, Languages and their Relation, Recursive and Recursively Enumerable Sets, Languages and Automata, Chomsky hierarchy of Languages

REGULAR GRAMMARS : Regular Sets and Regular Grammars, Converting Regular Expressions to Regular Grammars, Converting Regular Grammars to Regular Expressions, Left Linear and Right Linear Regular Grammars

UNIT-IV

CONTEXT- FREE LANGUAGES : Ambiguity in CFG, Leftmost and rightmost derivations, Language of a CFG, Sentential forms, Applications of CFG, Pumping Lemma for CFG, Derivations Generated by a Grammar, Construction of Reduced Grammars, Elimination of null and unit productions, Normal Forms for CFG: Chomsky Normal Form

SIMPLIFICATION OF CONTEXT- FREE GRAMMARS : Construction of Reduced Grammars, Greibach Normal Form

Assistant Dean
Division of Academic Affairs
Lovely Professional University

UNIT-V

PUSHDOWN AUTOMATA AND PARSING : Description and Model of Pushdown Automata, Representation of PDA, Acceptance by PDA, Pushdown Automata: NDPDA and DPDA, Context free languages and PDA, Pushdown Automata and Context-Free Languages, Comparison of deterministic and non-deterministic versions, closure properties, LL (k) Grammars and its Properties, LR(k) Grammars and its Properties, PARSING: Top-Down and Bottom-Up Parsing

UNIT-VI

TURING MACHINES AND COMPLEXITY : Turing Machine Model, Representation of Turing Machines, Design of Turing Machines, The Model of Linear Bounded Automaton, Power of LBA, Variations of TM, Non-Deterministic Turing Machines, Halting Problem of Turing Machine, Post Correspondence Problem, Basic Concepts of Computability, Decidable and Undecidable languages, RECURSIVELY ENUMERABLE LANGUAGE, Computational Complexity: Measuring Time & Space Complexity Power of Linear Bounded Automaton Variations of Turing Machine Cellular automaton

RECOMMENDED READINGS:

1. THEORY OF COMPUTER SCIENCE: AUTOMATA, LANGUAGES & COMPUTATION BY K.L.P.MISHRA & N. CHANDRASEKARAN PRENTICE HALL
2. AUTOMATA, COMPUTABILITY AND COMPLEXITY: THEORY AND APPLICATIONS BY ELAINE RICH, PEARSON
3. INTRODUCTION TO AUTOMATA THEORY, LANGUAGES, AND COMPUTATION BY HOPCROFT, MOTWANI, ULLMAN, PEARSON
4. INTRODUCTION TO THE THEORY OF COMPUTATION BY MICHAEL SIPSER, CENGAGE LEARNING
5. THEORY OF COMPUTATION: A PROBLEM SOLVING APPROACH BY KAVI MAHESH, WILEY
6. INTRODUCTION TO FORMAL LANGUAGES, AUTOMATA THEORY AND COMPUTATION BY KAMALA KRITHIVASAN RAMA R. PEARSON
7. THEORY OF COMPUTATION BY RAJESH K. SHUKLA, CENGAGE LEARNING
8. AN INTRODUCTION TO AUTOMATA THEORY AND FORMAL LANGUAGES. BY ADESH K. PANDEY, S.K. KATARIA & SONS
9. INTRODUCTION TO THEORY OF AUTOMATA, FORMAL LANGUAGES AND COMPUTATION BY SATINDER SINGH CHAHAL, GULJEET KAUR CHAHAL, A.B.S.PUBLICATION, JALANDHAR
10. AN INTRODUCTIONTO FORMAL LANGUAGES AND AUTOMATA BY PETER LINZ, JONES & BARTLETT LEARNING
11. CELLULAR AUTOMATA MACHINES: A NEW ENVIRONMENT FOR MODELING BY TOMMASO TOFFOLI MIT PRESS

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE332: INDUSTRY ETHICS AND LEGAL ISSUES

L: 2 T: 0 P: 0 Credits: 2

COURSE OUTCOMES: Through this course students should be able to

CO1 :: describe various types of funding and schemes for startup business in IT sector

CO2 :: observe the latest trends and current issues in information technology

CO3 :: observe ethical environment of work

CO4 :: analyze various cyber laws and related semantics for deploying security mechanisms in IT industry

CO5 :: describe various types of legal, ethical and professional issues in Information Security

CO6 :: distinguish types of intellectual properties.

UNIT-I

Ethics : Definition of ethics, Importance of integrity, Ethics in business world, improving corporate ethics, Ethical decision making, Ethics in information technology, Ethical behavior of IT professional, Supporting the ethical practices of IT users, reasons for ethical problems in business, Wealth, values, and human nature

UNIT-II

Companies : Introduction to IT and ITES industry (Product based, Services based), Introduction to NASSCOM, STPI, Overview on latest IT projects with global impact, Case study of an IT industry (Product based and Services based), Recent technology advancement, Current affairs related with the IT industry, Diversity in the Workforce

UNIT-III

Government Funding and schemes for Startup : what are Startups, Startup India benefits, Resources, bank loan for startup business, Start-up India, 10000 startups -A NASSCOM Initiative, Export promotion schemes: Software Technology Parks (STPs), Special Economic Zones (SEZ) Scheme Laws for Startups

UNIT-IV

Startup in IT : Planning of startup business in IT sector-Executive summary, General company Description, Products and services, Marketing plan, Operational plan, Management and organization, Personal Financial Statement, Startup Expenses and Capitalization, Financial Plan, Appendices, Refining the Plan Examples of Successful Start-ups

UNIT-V

Intellectual Properties (IP'S) : Concept of Intellectual Property, Copyright and Trademark, Different kinds of marks (brand names, logos, signatures, symbols, well known marks, certification marks and service marks Patents Importance of Patent Information in Business Development

UNIT-VI

Legal, Ethical and Professional issues in Information Security : Introduction, Law and Ethics in Information Security, Organizational Liability and the Need for Counsel, Policy Versus Law, Cyber Crime, Cyber-crime on the rise, Cyber law of India, Need for cyber law in India

RECOMMENDED READINGS:

1. ETHICS IN INFORMATION TECHNOLOGY BY GEORGE W REYNOLDS, CENGAGE LEARNING

CSE443: SEMINAR ON SUMMER TRAINING

L: 0 T: 0 P: 6 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: understand the area of interest and selection of an area of specialization.

CO2 :: explain their learning experience in training in the form of an internship report

CO3 :: examine their self-confidence and helps in finding their own proficiency

CO4 :: use meaningful and in-depth technical knowledge of training in their chosen field.

CO5 :: compose new learning from the training experience in the form of an oral N presentation

CO6 :: assess and enhance their employ ability skills and become job ready along with real corporate exposure.

Description:

The purpose is to provide an opportunity as a trainee to seek, identify and further develop the appropriate level of professionalism while observing the professional dealings at respective industry/organization and learn the latest techniques and skills to build a strong foundation for their career growth

Course Weightages:

ATT	CA	MTP	ETP1	ETP2
0	0	0	100	0

Attendance Requirement: NA

Continuous Assessment (CA): NA

ETP Evaluation Parameters:

Parameter	Marks
Internship Q and A	20
Justification of Answers to the questions based on the Mini project	10
Knowledge and skills applied and gained	15
Oral presentation	15
Organization introduction	10
Quality of the Project	10
Report Formatting and Content Quality	10
Successful implementation of the technologies/techniques learned during summer training in a mini	10
Total	**100**

Applicability of R Grade: Yes

The conduct of the course shall be as per guidelines issued by the University for the given course.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

INT246: SOFT COMPUTING TECHNIQUES

L: 2 T: 0 P: 2 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: describe the soft computing techniques in building the intelligent machines.

CO2 :: explain different neural networks for classification and clustering problems.

CO3 :: use fuzzy logic and reasoning to handle uncertainty and solve engineering problems.

CO4 :: compare and contrast genetic algorithms and swarm intelligence for optimization problems.

CO5 :: justify the performance and time complexity of hybrid systems.

CO6 :: develop the optimal models using available soft computing tools to solve real world problems.

UNIT-I

Introduction : artificial intelligence, artificial neural networks, genetic algorithms, swarm intelligent systems, expert systems

Neural network concepts : introduction to neural networks, biological neural networks to artificial neural networks, classification of neural networks, McCulloch-Pitts neuron model, learning rules, perceptron networks.

UNIT-II

Neural networks : back propagation neural networks, kohonen neural network, learning vector quantization, radial basis function neural networks, support vector machines.

UNIT-III

Fuzzy systems : basic definition and terminology, set-theoretic fuzzy operations, fuzzy sets and operations on fuzzy sets, fuzzy relations, fuzzy rules and fuzzy reasoning, fuzzy inference system, fuzzy based expert system.

UNIT-IV

Genetic algorithms : introduction to genetic algorithms, genetic operators, working of genetic algorithm, applications of genetic algorithm, genetic programming.

UNIT-V

Hybrid systems : hybrid systems, neuro-fuzzy systems, fuzzy genetic algorithms, neuro-genetics systems.

Swarm intelligence : swarm intelligence, cuckoo search, ant colony optimization, swarm intelligence in bees

UNIT-VI

Nature Inspired Optimization Techniques and Applications : Firefly Algorithm with application, Crow Search Algorithm, Hybrid Wolf-Bat Algorithm, Whale Search Algorithm, Moth-flame Optimization, Grasshopper Optimization

LABORATORY WORK:

List of Practical's

- Programs using Numpy, Pandas and Matplotlib
- Implementation of McCulloch Pitts neural network
- Implementation of Perceptron.
- Implementation of Back-propagation neural network

Assistant Dean
Division of Academic Affairs
Lovely Professional University

- Implementation of self-organizing map
- Implementation of linear vector quantization
- Implementation of radial basis network
- Implementation of fuzzy sets and operations on fuzzy sets.
- Implementation of fuzzy membership function.
- Implementation of fuzzy expert systems.
- Implementation of genetic algorithms.
- Implementation of neuro-fuzzy hybrid system.
- Implementation of neuro-genetic hybrid system.
- Implementation of artificial ant colony optimization.
- Implementation of artificial bee colony optimization.

RECOMMENDED READINGS:

1. SOFT COMPUTING WITH MATLAB PROGRAMMING BY NP PADHY , SP SIMON, OXFORD UNIVERSITY PRESS
2. PRINCIPLES OF SOFT COMPUTING BY S N SIVANANDAM, S N DEEPA, WILEY
3. NEURO-FUZZY AND SOFT COMPUTING: A COMPUTATIONAL APPROACH TO LEARNING AND MACHINE INTELLIGENCE BY JANG, SUN & MIZUTANI, PRENTICE HALL
4. NEURAL NETWORKS, FUZZY SYSTEMS AND EVOLUTIONARY ALGORITHMS : SYNTHESIS AND APPLICATIONS BY RAJASEKARAN, S., PAI, G. A. VIJAYALAKSHMI, PHI LEARNING

PEA305: ANALYTICAL SKILLS- I

L: 2 T: 1 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

- **CO1** :: reproduce the concepts learned to solve various questions of quantitative and reasoning aptitude
- **CO2** :: observe the data given and interpret it from the given problem
- **CO3** :: apply the Concepts to solve Company Specific Aptitude tests
- **CO4** :: analyze the problems and use logic to interpret and handle different situations
- **CO5** :: select the appropriate approach to initiate the given problem
- **CO6** :: solve quantitative and reasoning aptitude in competitive examinations

UNIT-I

Number system: HCF & LCM, divisibility rules, classification of numbers, factors, factorials, unit digit calculation, remainder properties

Average: basic average calculations, average increase and decrease, weighted average

A.P and G.P: basic formulas on arithmetic and geometric progressions, calculation of general terms, sum of n terms calculation

UNIT-II

Percentage: basic percentage calculations, percentage to fraction, percentage comparison, percentage increase and decrease, population change in percentage

Profit loss discount: basic concepts of cost price selling price and marked price, calculations of profit and loss percentage, types of discount and discount percentages, comparison of profit or loss with discount percentage

Simple and compound interest: basic concepts of interest calculations, comparison of simple and compound interest EMI calculations

UNIT-III

Logical reasoning: number series, alpha series, alphabet test, coding and decoding, language coding

UNIT-IV

Ratio and proportions: basic concepts of ratio and proportions and ages, problems based on ratio and proportions and ages problems based on partnerships and profit sharing

Alligation and mixtures: conceptual knowledge of alligation and mixtures, problems based on alligation and mixtures

UNIT-V

Permutation: basic principle of counting, numerical permutation (formation of numbers and the sum of numbers), alpha permutation(rearrangement of words and rank of a word), linear and circular permutation logical permutation

Combination: basic formulas of combination, formation of committee, combination of identical objects

Probability : the concept of probability, classification of events, conditional probability, problems based on coins dices, and cards

UNIT-VI

Analytical reasoning: Blood Relation, Direction Sense

Assistant Dean
Division of Academic Affairs
Lovely Professional University

RECOMMENDED READINGS:

1. QUANTITATIVE APTITUDE FOR COMPETITIVE EXAMINATIONS BY DR.R.S. AGGARWAL, S CHAND PUBLISHING
2. A MODERN APPROACH TO VERBAL AND NON-VERBAL REASONING BY DR. R.S. AGGARWAL S CHAND PUBLISHING
3. MAGICAL BOOK ON QUICKER MATHS BY M.TYRA, BANKING SERVICE CHRONICLE
4. MAGICAL BOOK SERIES ANALYTICAL REASONING BY M.K. PANDEY, BANKING SERVICE CHRONICLE

Assistant Dean
Division of Academic Affairs
Lovely Professional University

PEV107: VERBAL ABILITY-II

L: 1 T: 2 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: recall sentences to form paragraphs reflecting different patterns of organization by using distinct transition words.

CO2 :: examine the elements of an effective articulation in professional conversation

CO3 :: apply the learned strategies of skimming and scanning to discover the general idea and to find specific information in a familiar text.

CO4 :: analyze simple sentences containing learned vocabulary and using appropriate grammatical structures in speaking and writing

CO5 :: evaluate the different strategies to understand new vocabulary and grammatical structures in context.

CO6 :: compose grammatically structured questions related to basic needs and respond appropriately using short phrases and sentences.

UNIT-I

Sentence correction: modifiers, parallelism, subject-verb agreement, pronoun agreement, comparisons, redundancy, error of participles, verb tenses

UNIT-II

Voice and accent: introduction to vowels and consonants, introduction to syllable, stress and intonation

UNIT-III

Vocabulary enrichment: one- word substitution, Cloze test, Sentence Synonyms, Vocabulary with pictures

E-mail writing: purpose and functional role of e-mail, structural components of e-mail, do's and don'ts of e-mail writing, exercise based on e-mail writing scenarios

UNIT-IV

Essay writing: idea elaboration, writing an introduction, logical sequencing of ideas, generating points or supporting ideas and examples, concluding the essay

Reading comprehension passages: types of question- inference, main idea, supporting idea, assumption

UNIT-V

Narration: direct and indirect speech, conversion of one speech to another, key terminologies, rules of conversion, exercises based on conversion

Cover letter: key elements of cover letter, useful words and phrases for cover letter, format of cover letter, exercise based on cover letter writing scenarios

UNIT-VI

Critical reasoning: concepts - premise, assumption, conclusion, strengthening statement, weakening statement, types and patterns of questions, tips and tricks to understand and solve critical reasoning, indicators to identify basic concepts of critical reasoning

RECOMMENDED READINGS:

1. COLLINS COMMON ERRORS IN ENGLISH BY COLLINS DICTIONARIES, HARPERCOLLINS PUBLISHERS.
2. ESSENTIALS OF ENGLISH GRAMMAR BY BAUGH, L. SUE, MC GRAW HILL.
3. OXFORD LIVING GRAMMAR BY KEN PATERSON, MARK HARRISON, NORMAN COE, OXFORD UNIVERSITY PRESS.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CAP819: GAME TOOL-II

L: 0 T: 0 P: 3 Credits: 2

COURSE OUTCOMES: Through this course students should be able to

CO1 :: understand the fundamentals of game development

CO2 :: construct interactive games using C# and unity

CO3 :: use game programming languages, frameworks and game engines

CO4 :: demonstrate the process, and use mechanics for game development

LABORATORY WORK:

An Overview of Unity Engine

- The Project View
- The Hierarchy View
- The Inspector
- The Toolbar
- The Scene View
- The Game View
- The Animation and Animators Views
- The Profiler and Version Control
- Customizing the Editor
- Unity's Basic Concepts

Setting the Stage with Terrain

- Unity's Terrain Engine
- Customizing Terrain
- Lighting and Shadows
- Adding a Skybox and Distance Fog
- Adding Water to Your Terrain

Building Your Environment: Importing Basic Custom Assets

- Design First, Then Build
- Importing Textures for Widget's Terrain
- Importing Basic Meshes
- Setting Up Simple Shaders and Materials

Creating Characters

- Character Capabilities in Unity
- Importing Characters and Other Non-Static Meshes

Scripting in Unity

- Picking a Script Editor—or, "Do You Want Autocompletion with That?"
- Fundamentals of Scripting in Unity

Writing the Character and State Controller Scripts

- Setting It Up and Laying It Out
- A Simple Third-Person Controller

Hooking Up the Animations

- Animation in Unity
- Animation API
- Setting Up the PC's Animations
- Creating Animations Inside Unity

Assistant Dean
Division of Academic Affairs
Lovely Professional University

Using Triggers and Creating Environment Interactions

- Triggers and Collision
- Setting Up a Basic Trigger Object
- Setting Up Other Kinds of Triggers

Building Adversaries and AI

- Artificial Intelligence: Definitely Artificial, Not Much Intelligence
- Some Simple AI Guidelines
- A Simple Workflow
- Setting Up a Simple Enemy
- Hooking Up Widget's Attacks
- Rewarding the Player for a Job Well Done
- Spawning and Optimization

Designing the Game's GUI

- Basic Interface Theory
- Unity's GUI System
- A Custom Skin for Widget
- Setting Up the HUD
- A Sample Pop-Up Screen
- Adding Full-Screen Menus

Creating Lighting and Shadows

- Types of Lights
- Lighting the World
- Creating Shadows
- Other Light Effects

Using Particle Systems

- Particles: From Smoke to Stardust
- Setting Up a Simple System
- Particles for Widget

Adding Audio and Music

- Setting Up a Simple Audio Clip
- Ambient Sound Effects
- Controlling Sounds Through Scripts
- Adding Background Music

Creating the Final Build

- Prepping for the Build
- Other Build Features

RECOMMENDED READINGS:

1. GAME DEVELOPMENT WITH UNITY BY MICHELLE MENARD, CENGAGE LEARNING
2. HOLISTIC GAME DEVELOPMENT WITH UNITY BY PENNY DE BYL, FOCAL PRESS.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE423: VIRTUALIZATION AND CLOUD COMPUTING

L: 3 T: 0 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: illustrate the main concepts key technologies strengths of Virtualization technology

CO2 :: examine the appropriate technologies, algorithms, and approaches for provisions of cloud computing

CO3 :: formulate the core issues of cloud computing such as security, privacy, and interoperability

CO4 :: evaluate the appropriate cloud computing solutions and recommendations according to the applications used

CO5 :: enumerate the emerging areas of cloud computing and how it relates to traditional models of computing

CO6 :: understand the service attributes for compliance with enterprise objectives

UNIT-I

Virtualization techniques : virtualization technology, overview of x86 virtualization, types of virtualization concept of VLAN SLAN and VSAN and benefits concept of VLAN VSAN and benefits

Overview of Distributed computing : Parallel and Distributed Systems, Parallel Computing, Parallel Computer Architecture, Distributed Systems, Differences and Similarities among Different Types of Computing

UNIT-II

Introduction to Cloud Computing : Cloud Computing in a Nutshell, Roots of Cloud Computing., Layers and Types of Clouds., Desired Features of a Cloud, Cloud Infrastructure Management., Examining the Characteristics of Cloud Computing, cloud types

Migrating into a Cloud : Broad Approaches to Migrating into the Cloud, The Seven-Step Model of Migration into a Cloud VM Migration, Cloud Middleware and Best Practices, Concept and Need of Cloud Middleware QoS Issues in Cloud Data Migration and Streaming in Cloud Interoperability

UNIT-III

Understanding cloud architecture : exploring the cloud computing stack, Workload distribution architecture, Capacity planning, Cloud bursting architecture, Disk provisioning architecture, Dynamic failure detection and recovery architecture

Cloud mechanisms : Automated scaling listerner, Load balancer, Pay-per-use monitor, Audit monitor, SLA monitor, Fail-over Systems, Resource Cluster, Service level agreements, Service- Oriented Architecture

UNIT-IV

Cloud Computing Technologies and Applications : Model Application Methodology, Cloud-Based High Performance Computing Clusters, Diving into Infrastructure as a Service, Exploring PaaS, Understanding SaaS Different Cloud Providers and Comparison of Services Resource management

Cloud Economics : Developing an Economic Strategy, Exploring the Costs, Laws of cloudonomics, Cost estimation

Assistant Dean
Division of Academic Affairs
Lovely Professional University

UNIT-V

Cloud security : Risks and threats, Security mechanisms, Vulnerability checklist, Establishing a Cloud Governance Strategy, Managing Service Levels, Developing a Secure, Accountable, Reliable Cloud Environment

Managing cloud storage : Storage Models, File Systems, Distributed File Systems, General Parallel File System, backup techniques, Principles of Workload Management, Improving Security and Governance

UNIT-VI

Container technology : Introduction to containers, container architectures, Docker containers, Kubernetes

Cloud Platforms in Industry : Amazon Web services, Google App Engine, Microsoft Azure, Case studies

Other aspects of Cloud : Edge Computing, Fog Computing, IIoT, Green Cloud computing practices, Complexity in Cloud-native systems

RECOMMENDED READINGS:

1. CLOUD COMPUTING: FUNDAMENTALS, INDUSTRY APPROACH AND TRENDS BY RISHABH SHARMA, WILEY
2. CLOUD COMPUTING: PRINCIPLES AND PARADIGMS BY RAJKUMAR BUYYA, JAMES BROBERG, ANDRZEJ GOSCINSKI, WILEY
3. MASTERING CLOUD COMPUTING BY RAJ KUMAR BUYYA, CHRISTIAN VECCHIOLA,S. THAMARAI SELVI, MC GRAW HILL

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE427: VIRTUALIZATION AND CLOUD COMPUTING LABORATORY

L: 0 T: 0 P: 2 Credits: 1

COURSE OUTCOMES: Through this course students should be able to

CO1 :: analyze key technologies and capabilities required for setting up IT virtualization and cloud computing infrastructure

CO2 :: examine the ultimate goal of assessing, measuring and planning for the deployment of cloud-based IT resources

CO3 :: illustrate the knowledge of cloud computing technology architectures based on Software as- a-Service (SaaS), Platform-as-a-Service (PaaS) and Infrastructure-as-a-Service (IaaS) delivery models.

CO4 :: describe the applications of tools used in Cloud environment.

CO5 :: assess the real life use of Storage and Network facilities in virtualized infrastructure.

CO6 :: recognize the difference in the working of various virtualization tools.

LABORATORY WORK:

Understanding virtualization

- Virtualization and Cloud Computing
- Virtualizing servers
- Virtualizing desktops
- Virtualizing applications
- BIOS setting of Physical machine for virtualization technology

Understanding hypervisors

- Exploring the hypervisors
- Understanding type 1 hypervisor
- Understanding type 2 hypervisor
- Resource allocation
- Vmware ESX

Understanding virtual machines

- Examining CPU's in a virtual machine
- Examining memory in a virtual machine
- Examining network resources in a virtual machine
- Examining storage in a virtual machine
- Understanding how a virtual machine works
- Understanding virtual machine clones
- Understanding templates
- Understanding snapshots
- Understanding OVF
- Creating a virtual machine
- VM configuration
- Full and Linked Clone in VMware Workstation

- Exploring VMware Workstation
- Installation of VMware Workstation
- Installing a guest OS
- Installing windows on a virtual machine
- Loading windows into a virtual machine
- Installing vm ware tools
- Understanding configuration options
- Optimizing a new virtual machine
- Installing linux on a virtual machine

Management With v Center Server

- V Center 6 Overview
- Creating a Virtual Machine in HOL
- Cloning VMs and using Templates
- Tagging and Search to find objects quickly
- Monitoring events and creating alarms
- Migrating VMs with VMware vMotion
- vSphere Monitoring and Performance

Introduction to vSphere Network and Security

- Adding and Configuring vSphere Standard Switch
- Adding and Configuring vSphere Distributed Switch
- User Access and Authentication Roles
- Understanding Single Sign On
- Introduction to vSphere Storage
- Sphere Storage Overview
- Creating and configuring vSphere Datastores
- Capacity and Cost Management (vRealize Operations)
- Optimize capacity with what-if scenarios and costs
- View Costs in vRealize Operations

Monitoring Applications and Services

- Configure Service Discovery configuration
- Service Monitoring
- Service Discovery
- Application Monitoring

Automation Central and Preparing for Cloud/Hybrid Data center

- Creating a job from VM
- Recurring Jobs
- What-if analysis to VMware Cloud on AWS Service

Container technology

- installation
- Working with containers
- Configuring containers
- Building web server

RECOMMENDED READINGS:

1. CLOUD COMPUTING BIBLE BY BARRIE SOSINSKY, WILEY

INT247: MACHINE LEARNING FOUNDATION

L: 2 T: 0 P: 2 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: describe the concepts of classification and regression algorithms.

CO2 :: examine meaningful features from a given dataset by learning pre-processing skills

CO3 :: apply the validated machine learning models in a given situation for an available dataset

CO4 :: identify the dimensionality reduction using lda, pca and kpca.

CO5 :: evaluate the problem that categorize into supervised, unsupervised and reinforcement learning

CO6 :: develop a machine learning model to solve a real world problem

UNIT-I

Giving computers the ability to learn from data: making predictions about the future with supervised learning, solving interactive problems with reinforcement learning, discovering hidden structures with unsupervised learning, roadmap for building machine learning systems, using Python for machine learning

Building good training sets: data preprocessing, dealing with missing data, handling categorical data, partitioning a dataset in training and test sets, normalization, selecting meaningful features

UNIT-II

Machine learning classifiers using scikit-learn: choosing a classification algorithm, first steps with scikit-learn, modeling class probabilities via logistic regression, maximum margin classification with support vector machine, decision tree learning, k-nearest neighbor algorithm, bayesian learning, majority voting classifier, bagging and boosting classifier, random forest classifier

UNIT-III

Predicting continuous target variables with regression analysis: introducing linear regression, relationship using a correlation matrix, exploratory data analysis, regularized methods for regression, polynomial regression, modeling nonlinear relationships in the housing dataset, decision tree and random forest regressor

UNIT-IV

Working with unlabeled data: K-means clustering, hard versus soft clustering, using the elbow method to find the optimal number of clusters, silhouette plots, organizing clusters as a hierarchical tree, agglomerative clustering, DBSCAN clustering

UNIT-V

Dimension reduction: unsupervised dimensionality reduction via principal component analysis, supervised data compression via linear discriminant analysis, using kernel principle component analysis for nonlinear mappings, projecting new data points

UNIT-VI

Model evaluation and hyperparameter tuning: streamlining workflows with pipelines, using k- fold cross validation to access model performance, debugging algorithms with learning and validation curves, fine-tuning machine learning models via grid search

Assistant Dean
Division of Academic Affairs
Lovely Professional University

LABORATORY WORK:

List of Practical's:

<table>
<tr><th>#</th><th>Description</th></tr>
<tr><td>1</td><td>Given the following vectors:
A = [1,2, 3, 4, 5, 6, 7, 8, 9, 10]
B = [4, 8, 12, 16, 20, 24, 28, 32, 36, 40]
C = [10, 9, 8, 7, 6, 5, 4, 3, 2, 1]

Ex. 1: Find the arithmetic mean of vector A, B and C
Ex. 2: Find the variance of the vector A, B and C
Ex. 3: Find the Euclidean distance between vector A and B
Ex. 4: Find the correlation between vectors A & B and A & C</td></tr>
<tr><td>2</td><td>Load breast cancer dataset and perform classification using Euclidean distance. Use 70% data as training and 30% for testing.</td></tr>
<tr><td>3</td><td>Repeat the above experiment with 10-fold cross validation and find the standard deviation in accuracy.</td></tr>
<tr><td>4</td><td>Repeat the experiment 2 and build the confusion matrix. Also derive Precision, Recall and Specificity of the algorithm.</td></tr>
<tr><td>5</td><td>Predict the class for **X = < Sunny, Cool, High, Strong >** using Naive Bayes Classifier for given data $P(C \mid X) = \frac{P(X \mid C).\, P(C)}{P(X)}$
<table>
<tr><th>#</th><th>Outlook</th><th>Temp</th><th>Humidity</th><th>Windy</th><th>Play</th></tr>
<tr><td>D1</td><td>Sunny</td><td>Hot</td><td>High</td><td>False</td><td>No</td></tr>
<tr><td>D2</td><td>Sunny</td><td>Hot</td><td>High</td><td>True</td><td>No</td></tr>
<tr><td>D3</td><td>Overcast</td><td>Hot</td><td>High</td><td>False</td><td>Yes</td></tr>
<tr><td>D4</td><td>Rainy</td><td>Mild</td><td>High</td><td>False</td><td>Yes</td></tr>
<tr><td>D5</td><td>Rainy</td><td>Cool</td><td>Normal</td><td>False</td><td>Yes</td></tr>
<tr><td>D6</td><td>Rainy</td><td>Cool</td><td>Normal</td><td>True</td><td>No</td></tr>
<tr><td>D7</td><td>Overcast</td><td>Cool</td><td>Normal</td><td>True</td><td>Yes</td></tr>
<tr><td>D8</td><td>Sunny</td><td>Mild</td><td>High</td><td>False</td><td>No</td></tr>
<tr><td>D9</td><td>Sunny</td><td>Cool</td><td>Normal</td><td>False</td><td>No</td></tr>
<tr><td>D10</td><td>Rainy</td><td>Mild</td><td>Normal</td><td>False</td><td>Yes</td></tr>
<tr><td>D11</td><td>Sunny</td><td>Mild</td><td>Normal</td><td>True</td><td>Yes</td></tr>
<tr><td>D12</td><td>Overcast</td><td>Mild</td><td>High</td><td>True</td><td>Yes</td></tr>
<tr><td>D13</td><td>Overcast</td><td>Hot</td><td>Normal</td><td>False</td><td>Yes</td></tr>
<tr><td>D14</td><td>Rainy</td><td>Mild</td><td>High</td><td>True</td><td>No</td></tr>
</table>
Ans: Label = NO</td></tr>
<tr><td>6</td><td>For the data given in Exercise 5, find the splitting attribute at first level:
Information Gain: $I(P, N) = -\frac{P}{S}\log_2\frac{P}{S} - \frac{N}{S}\log_2\frac{N}{S} = 0.940$
$Entropy: E(Outlook) = \sum_{i=1}^{v} \frac{P_i + N_i}{P + N} I(P_i, N_i) = 0.694$
$Gain\ (Outlook) = I(P, N) - E(Outlook) = 0.246$</td></tr>
</table>

Assistant Dean
Division of Academic Affairs
Lovely Professional University

Ans:

Attribute	Gain
Outlook	**0.246**
Temperature	0.029
Humidity	0.151
Windy	0.048

7 Generate and test decision tree for the dataset in exercise 5

8 Find the clusters for following data with k = 2: Start with points1 and 4 as two separate clusters.

i	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Ans:

i	**Point**
C_1	1, 2
C_2	3, 4, 5, 6, 7

9 Find following statistics for the data given in Exercise 1

$$Within\ Class\ Scatter: S_W = \sum_{i=1}^{c} \sum_{x \in w_i} (x - m_i)(x - m_i)^T$$

$$Between\ Class\ Scatter: S_B = \sum_{i=1}^{c} n_i (m_i - m)(m_i - m)^T$$

$$Total\ Scatter: S_T = \sum_{i=1}^{M} (x_i - m)(x_i - m)^T$$

10 Given the following vectors:

X = [340, 230, 405, 325, 280, 195, 265, 300, 350, 310]; %sale

Y = [71, 65, 83, 74, 67, 56, 57, 78, 84, 65];

Ex. 1: Find the Linear Regression model for independent variable X and dependent variable Y.

Ex. 2: Predict the value of y for x = 250. Also find the residual for y_4.

RECOMMENDED READINGS:

1. PYTHON MACHINE LEARNING by SEBASTIAN RASCHKA, PACKT PUBLISHING
2. LEARNING SCIKIT-LEARN: MACHINE LEARNING IN PYTHON by RAÚL GARRETA, GUILLERMO MONCECCHI, PACKT PUBLISHING

Assistant Dean
Division of Academic Affairs
Lovely Professional University

INT417: MACHINE LEARNING ALGORITHMS

L: 3 T: 0 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: categorize the machine learning problems based on learning rules.

CO2 :: apply the key concepts that form the core of machine learning.

CO3 :: develop the key algorithms for the system that are intelligent enough to make the decisions.

CO4 :: contrast the statistical, computational and game-theoretic models for learning.

UNIT-I

Introduction to machine learning : learning, need of machine learning, types of learning, wellposed learning problems, designing a learning systems, issues in machine learning

Formal learning model : statistical learning framework, empirical risk minimization, empirical risk minimization with inductive bias, PAC learning, general learning model

UNIT-II

Decision tree learning : decision tree representation, appropriate problem for decision tree learning, CART, ID3, C4.5, information gain measure, inductive bias in decision tree, minimum description length, Issues in decision tree learning, random forest

UNIT-III

Generative models : maximum likelihood estimator, naive Bayes, EM Algorithm, Bayesian learning, Bayes theorem, Bayesian decision theory, brute-force concept learning, minimum error rate classification, discriminant function, Bayes optimal classifier, gibs algorithm, Bayesian belief network

UNIT-IV

Complex prediction problems : one-versus-all, all-pairs, linear multiclass predictors, multiclass SVM and SGD, structured output predictions, ranking, multivariate performance measure

Nonparametric Methods : Density estimation, parzen window, the nearest neighbour rule, KNearest neighbour estimation.

UNIT-V

The bias-complexity tradeoff : no free lunch theorem, error decomposition, the VC-Dimension, surrogate loss functions, the Rademacher complexity, the Natarajan dimension

Nonuniform learnability : learning via uniform convergence, characterizing nonuniform learnability, structural risk minimization, minimum description length, different notions of learnability, computational complexity of learning

UNIT-VI

Algorithm-Independent machine Learning : combining classifiers, re-sampling for estimating statistics, lack of inherent superiority of classifier, comparing and estimating classifiers

Reinforcement learning : the learning task, Q learning, nondeterministic rewards and action, temporal difference learning

RECOMMENDED READINGS:

1. MACHINE LEARNING BY THOMAS MITCHELL, MCGRAW HILL EDUCATION
2. PATTERN CLASSIFICATION BY RICHARD O. DUDA, PETER E. HART, DAVID G. STORK, WILEY

PEA306: ANALYTICAL SKILLS-II

L: 2 T: 1 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: reproduce the concepts learned to solve various questions of quantitative and reasoning aptitude

CO2 :: observe the data given and interpret it from the given problem

CO3 :: apply the Concepts to solve Company Specific Aptitude tests

CO4 :: analyze the problems and use logic to interpret and handle different situation

CO5 :: select the appropriate approach to initiate the given problem

CO6 :: solve quantitative and reasoning aptitude in competitive examinations

UNIT-I

Time and Work: problems based on time and work, formulae, computation of work done together, efficiency based problems, men, women, and children based problems, wages based work problems.

Pipes and Cisterns: inlet-outlet,part of tank filled, time based problems.

UNIT-II

Time, speed and distance: concept of time, speed and distance, conversion of units and proportionality, average speed concept.

Problem on trains: relative speed concept & application

Boats and Streams: down stream and upstream

UNIT-III

Syllogism: logical venn diagrams, ability to relate a certain given group diagrammatically

Arithmetical reasoning: venn diagrams based problems

Number ranking test: number test and ranking test problems

UNIT-IV

Mensuration: problems on surface area and volume of cube, cuboids, problems on surface area and volume of sphere and hemisphere, problems on surface area and volume of cone and cylinder.

Height and distance: problembased on height and distance.

UNIT-V

Calendar: basic conceptof calendar,date and days, finding the exact day.

Clocks: concept of clock, facts and formulae, practice problems.

Analytical Reasoning: Linear seating arrangement, circular seating arrangement, puzzle test.

UNIT-VI

Data interpretation: basics of data interpretation and its types, tabulation, bar graph, pie chart, and line graph.

Data sufficiency: check sufficiency of data.

RECOMMENDED READINGS:

1. A MODERN APPROACH TO VERBAL & NON-VERBAL REASONING BY DR. R. S. AGGARWAL, S CHAND PUBLISHING.
2. QUANTITATIVE APTITUDE BY DR. R. S. AGGARWAL, S CHAND PUBLISHING.
3. ANALYTICAL REASONING BY MK PANDAY, BANKING SERVICE CHRONICLE.
4. MAGICAL BOOK ON QUICKER MATHS BY M.TYRA, BANKING SERVICE CHRONICLE.

PES319: SOFT SKILLS-II

L: 1 T: 2 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: understand the professional attitude and its application in building a self-brand

CO2 :: demonstrate confidence and leadership qualities

CO3 :: analyze of job profiles in the light of career opportunities

CO4 :: articulate effectively in the interview

CO5 :: develop communication skills with guided practice

CO6 :: classify professional ethics to be followed

UNIT-I

Professional attitude and branding: introduction to professional attitude, tips to create a professional attitude in the workforce, importance of positive attitude at the workplace, ways to do professional branding, dress for D-Day, creating an impressive profile on LinkedIn, tips for virtual meeting etiquette

UNIT-II

Resume building: the difference between a resume and CV, types of resumes (1page resume, detailed resume, and digital resume), 7 components to a resume, resume writing tips, video resume making, tips for creating a video resume(script, filming space, recording device, additional visuals)

UNIT-III

Communication and presentation skills: introduction to communication, barriers to communication, introduction to KYC, the importance of knowing the company, discussion on various KYCs of product and service-based companies, important aspects to consider in KYC.

UNIT-IV

Group discussion: recap of the concept of group discussions, types of group discussion, techniques to generate ideas - SPELT, KWA, 5Ws 1H, brainstorming, POPBEANS, VAP, SCAMPER, do's and don'ts of group discussion tips to crack virtual group discussion

UNIT-V

Interviews: pre-placement talks, pre-interview preparation, interview preparation, re-cap of power dressing, interview etiquette in online and offline scenarios, types of interviews, various answering techniques virtual interviews do's and don'ts of virtual interviews.

UNIT-VI

Work culture etiquette: etiquette in the workplace, time management, decision-making skills, email writing, do's and don'ts of email writing.

RECOMMENDED READINGS:

1. SOFT SKILLS: KNOW YOURSELF AND KNOW THE WORLD BY DR. K. ALEX, S CHAND PUBLISHING
2. PERSONALITY DEVELOPMENT AND SOFT SKILLS BY BARUN K. MITRA, OXFORD UNIVERSITY PRESS
3. THE ACE OF SOFT SKILLS: ATTITUDE, COMMUNICATION AND ETIQUETTE FOR SUCCESS BY GOPALASWAMY RAMESH AND MAHADEVAN RAMESH, PEARSON

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CAP820: GAME TOOL-III

L: 0 T: 0 P: 3 Credits: 2

COURSE OUTCOMES: Through this course students should be able to

CO1 :: define the different types of games and its concepts

CO2 :: enumerate the game development using various animation tools

CO3 :: understand the functioning of designed game in unreal engine

CO4 :: apply game designing fundamental to develop game using unreal engine

LABORATORY WORK:

Fundamental of Game Design in Unreal

- Installation of Unreal Engine
- Game Development basics
- Unreal Fundamentals

Inputs, Collisions, and Blueprints

- Exporting and importing of game elements in Unreal
- Inputs in Unreal, Blueprints and Optimization
- Use of Nanite in Unreal Engine

Concept of Lighting and Materials

- Fundamentals of lighting
- Fixed lighting in Unreal
- Dynamic lighting and other lighting effects in Unreal
- Concept of Lumen
- Working with Materials
- Altering Material Appearance

Implementation of Physics in Unreal

- Introduction to Physics
- Constraints Motors and Forces
- Additional Physics Applications

Use of Blueprints in Unreal

- Introduction to Blueprints
- Blueprint Concepts
- Optimization and Summary
- Blueprint Casting
- Blueprint Interfaces

Landscapes and Skeletal Meshes in Unreal

- Introduction to Landscapes
- Landscape Materials and Foliage
- Introduction to Skeletal Meshes

Audio and Artificial Intelligence in Unreal

- Introduction to Audio
- Supplemental Audio Techniques
- Introduction to Artificial Intelligence
- Logic and Behavior Trees

Assistant Dean
Division of Academic Affairs
Lovely Professional University

Publishing and Optimization of Unreal Games

- Preparing for Publishing
- Publishing with Unreal
- Optimization of Unreal games
- Animating with Skeletal Meshes

RECOMMENDED READINGS:

1. UNITY 2018 GAME DEVELOPMENT IN 24 HOURS BY MIKE GEIG, PEARSON.
2. FUNDAMENTALS OF GAME DESIGN BY ADAMS, PEARSON

Assistant Dean
Division of Academic Affairs
Lovely Professional University

INT248: ADVANCED MACHINE LEARNING

L: 2 T: 0 P: 2 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: define and select various constructors, functions, parameters and their types in the implementation of different neural networks using tensor flow.

CO2 :: describe and compare the different models for solving the problems of regression and classification using keras tens flow.

CO3 :: apply deep neural networks model to solve the classification problem of images by proper planning .

CO4 :: develop and select various constructors, functions, parameters and their types in the implementation of auto encoder based problems.

CO5 :: analyze the model of the deep neural networks to solve the problem of sequential data using recurrent neural network approach and evaluate its performance.

CO6 :: categorize the models of deep neural networks to solve the problem of sequential data using generative adversarial neural network approach and evaluate its performance.

UNIT-I

Building models with Tensor Flow : Introduction to Tensor Flow, Installation of Tensor Flow, Tensor Flow ranks and tensors, Tensor Flow's computation graphs, variables in Tensor Flow, building a regression model, multi-layer Perceptron learning for classification, tensor flow optimizers, transforming tensors as multidimensional data arrays

UNIT-II

Building models with Keras : Introduction to keras, Keras installation, keras layers and models, image classification with keras, building text classification model, implementing linear regression model, over fit and under fit, save and load model, hyper parameter tuning

UNIT-III

Classifying images with deep convolutional neural networks : building blocks of convolutional neural networks, determining the size of the convolution output, performing a discrete convolution in 2D, subsampling, putting everything together to build a CNN, Implementing a deep convolutional neural network using Tensor Flow, Transfer learning with pre-trained CNN, Data Augmentation, Image segmentation

UNIT-IV

Auto encoders and Pre-trained CNN : Introduction to auto encoders, need for auto encoders, architecture of auto encoder, properties and hyper parameter, types of auto encoders, data compression using auto encoders, variational auto encoders

UNIT-V

Modelling sequential data using recurrent neural networks : modelling sequential data, understanding the structure and flow of an RNN, computing activation in an RNN, challenges of learning long-range interactions, implementing a multilayer RNN for sequence modelling in Tensor Flow, text classification with an RNN, text generation with an RNN, time series forecasting, LSTM units, sequence classification with LSTM, stacked LSTM for sequence classification

Assistant Dean
Division of Academic Affairs
Lovely Professional University

UNIT-VI

Generative Adversarial Networks: Introduction to generative models, overview of GAN structure, discriminator, generator, building GAN, problems with GANs, Cycle GAN

LABORATORY WORK:

Parallelizing neural network training with Tensor Flow

- classifying handwritten digits on MNIST dataset
- developing a simple model with the low-level Tensor Flow API
- training neural networks efficiently with high-level Tensor Flow APIs
- developing a multilayer neural network with Keras

The mechanics of Tensor Flow

- Tensor Flow's computation graphs
- building a regression model
- transforming tensors as multidimensional data arrays
- visualizing the graph with Tensor Board

RECOMMENDED READINGS:

1. PYTHON MACHINE LEARNING by SEBASTIAN RASCHKA, VAHID MIRJALILI, PACKT PUBLISHING
2. ADVANCED MACHINE LEARNING WITH PYTHON by JOHN HEARTY, PACKT PUBLISHING

Assistant Dean
Division of Academic Affairs
Lovely Professional University

INT407: INFORMATION SECURITY AND PRIVACY

L: 3 T: 0 P: 0 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: define the security controls sufficient to provide a required level of confidentiality, integrity and availability in an organization's computer systems

CO2 :: explore the key technical tools available for security/privacy protection

CO3 :: discover vulnerabilities critical to the information assets of an organization

CO4 :: differentiate unethical and illegal behavior in information security

CO5 :: review major standards that relate to the practice of information security

CO6 :: understand authentication methods by using biometrics as an authentication method

UNIT-I

Information Systems: Information system security & threats, meaning and importance of information systems, information security and privacy threat

Building Blocks of Information Security: principles, terms and three pillars of information security, risk management & risk analysis, information classification, approaches and considerations for risk analysis

UNIT-II

Threats: new technologies open door threats, level of threats: information, network Level, classifications of threats and assessing damages

Program security: overview of program security, types of flaws, viruses and other malicious code, controls against program threats covert channels

UNIT-III

Biometrics Controls for Security: access control, user identification & authentication, biometric techniques, face recognition and related issues, key success factors, advanced minutiae based algo

UNIT-IV

Security standards and policies: intro to ISO 27001. COBIT, SSE-CMM, policies and their elements, HIPAA security guidelines, methodologies for information system security.: IAM, IEM, SIPES.

UNIT-V

Security metrics and trusted system: Security matrix, Classification, Privacy vs security, Security Models Trusted OS Design and Principles Security features of Trusted OS

UNIT-VI

Privacy Technological Impacts: Impact of information technology on privacy of an individual, Affect of web technologies on privacy, RFID related privacy issues, Internet related privacy issues

RECOMMENDED READINGS:

1. INFORMATION SYSTEMS SECURITY WILEY PUBLICATIONS BY NINA GODOLE, WILEY
2. NETWORK SECURITY: THE COMPLETE REFERENCE ROBERTA: TATAMCGRAW HILL BY BRAGG MCGRAW HILL EDUCATION

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CAP821: PROJECT

L: 0 T: 0 P: 4 Credits: 2

COURSE OUTCOMES: Through this course students should be able to

CO1 :: learn the skills required for working at different positions in software project

CO2 :: understand the knowledge, skills and attitudes of a professionals

CO3 :: apply the knowledge and skills gained during degree program to generate new knowledge

CO4 :: demonstrate rigorous methods to solve problems related to a substantive area of the study

Description:

A project involves activities that closely relates to professional work in the related field. Upon completion, student will be able to apply the discipline knowledge and capabilities acquired during the programme. A capstone project may incorporate work-integrated learning placements, case analysis and other immersive learning experiences such as study tools, service learning, volunteering, virtual simulations, etc.

Course Weightages:

ATT	CA	MTP	ETP1	ETP2
0	50	0	50	0

Attendance Requirement: Weekly as per time table

Continuous Assessment (CA): Monthly by the supervisor

ETP Evaluation Parameters:

Parameter	Marks
Attendance of the student (* If Att% 76-100 Award:4-5 Marks , 51-75% Award:3 Marks, 26-50% Award:2 Marks, 0-25% Award: 1 Mark)	5
Plan of work was well defined and was realistic to be achieved in stipulated timelines	2
Student has done feasibility analysis required for completion of Project Work such as cost issues, time, operational feasibility etc.	2
Student was able to narrow down the title based upon the preliminary survey.	4
Testing of the project has been successfully done by the student	3
The categorization of chapters was appropriate and relevant	2
The flow of the written report was well organized and structured.	2

Assistant Dean
Division of Academic Affairs
Lovely Professional University

The plagiarism of the report was as per the prescribed limit.	2
The project has been completed and is in presentable form	3
The relevant risks related to the project have been considered.	1
The report was well written and was free from grammatical errors	2
The results obtained by the student are encouraging and supports the proposed work	4
The steps/ action plan/ blueprint/ algorithm/ Methodology formulated by the student is appropriate to solve the problem	4
The student has completed the real implementation of the project work	4
The student has explored some project topics relevant to his/her field/area.	2
The student has explored some real life problems related to which project topic can be finalized.	3
The student has gathered all the relevant tools/ resources/ material required for completion of the project in stipulated timelines	3
The student has identified all the resources/tools/ equipment's/ material required for completion of project work till date.	2
The student has started the real implementation of the project work	3
The student has sufficient knowledge regarding the tools/equipment's to be used in the project.	3
The student was able to overcome/ handle the problems/ challenges encountere during implementation	3
The title was clear and comprehensive rather than broad and general	3
Title proposed by the student justifies the scope of the proposed work	3
Total	65

ETP Evaluation Parameters:

Parameter	Marks
Adequate references/bibliography are included in the report and are in proper format.	3

Assistant Dean
Division of Academic Affairs
Lovely Professional University

The content mentioned in the report is Accurate and free from errors/ grammatical mistakes.	3
The plagiarism in the final report is as per the prescribed policy	3
All the required data has been collected and analysed properly	3
All the results obtained from experimentation/ survey/ review/ field work are significant and supports the project work	3
The candidate has been able to generate any of the following as a outcome a) IPR generation (Patents/ Copyrights/ design/ prototype), b) New innovations, c) Quality publications, d) Societal Contributions, e) Policy Making	8
The candidate is confident while answering the questions and has an ability to remain calm under pressure.	3
The candidate is open to the observations and pertinent suggestions formulated during the presentation/ discussion	2
The candidate is well updated about the recent developments and trends in the proposed area of research	3
The introductory pages and annexures (list of tables, figures, abstract, index, ethical committee approvals, published materials, COA etc.) are included in the written report.	3
The overall formatting of the report is as per the prescribed guidelines	3
The presentation includes the relevant information and is precise, correct and comprehensive	2
The Project work has addressed any of the following a) Societal Issues, b) Industrial issues, c) Government issues (Policy making etc.),d) Business issue, e) Environmental issues, f) Conceptual, g) Enrich the discipline, h) News worthy	4
The project work will solve the problem at any of the following a) organization, b)State level, c) National Level, d) International Level	3

Assistant Dean
Division of Academic Affairs
Lovely Professional University

The student was able to achieve all the formulated objectives of the proposed work	4
Total	**50**

Applicability of R Grade: Yes

The conduct of the course shall be as per guidelines issued by the University for the given course.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE434: GAME DEVELOPMENT IN 3D

L: 2 T: 0 P: 2 Credits: 3

COURSE OUTCOMES: Through this course students should be able to

CO1 :: apply 3D modelling concept for game development process

CO2 :: develop a game with different themes and levels by adding various control functions in it

CO3 :: write a game with different themes and levels by adding various control functions in it

CO4 :: analyze the various game components and materials for game play workflows

CO5 :: evaluate the user interface elements for game building process

CO6 :: construct and develop game design through story telling using unity 3D

UNIT-I

Working in Unity 3D : Working in 3D Space, Adding Behaviors, Working in Unity Particles(Materials, Lightening, Audio and Camera Positioning)

UNIT-II

Player Controls and Positioning : Motion Sensoring, Game Manager Physics, Gameplay components, Objects in 3 D game kit

UNIT-III

3D concepts for game play : Controller, Game art creation

UNIT-IV

Programming : Scripting language, Basics of C# language

UNIT-V

Gameplay components : Game creation in 3D, Adding Enemy to the game

UNIT-VI

Working with Navmesh : Basics of Navmesh, Path finding using Navmesh

LABORATORY WORK:

List of Practicals

- Working with Unity Interface
- Create a Game scene in unity 3D
- Use terrain to create game scene with effects
- Add 3DPhysics to Game scene
- Use Prefab to game scene
- Add first person and third person controller to game scene.
- Use of scripting language in game scene with AI concept

RECOMMENDED READINGS:

1. BUILDING A GAME WITH UNITY AND BLENDER BY LEE ZHI ENG BY LEE ZHI ENG, PACKT PUBLISHING

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE445: CAPSTONE PROJECT

L: 0 T: 0 P: 20 Credits: 10

COURSE OUTCOMES: Through this course students should be able to

CO1 :: integrate and synthesize prior knowledge and learning from multiple and diverse topic areas

CO2 :: develop openness to new ideas in computer science, develop the ability to draw reasonable inferences from observations and learn to formulate and solve new problems using analytical and problem-solving skills

CO3 :: demonstrate holistically the development of graduate capabilities

CO4 :: identify the intricacies involved in solution design to real world problems

CO5 :: devise solutions to real world problems which are capable of delivering unparalleled performance

CO6 :: illustrate and instil the importance of teamwork while building solutions to real world problems

Description:

A project involves activities that closely relates to professional work in the related field. Upon completion, student will be able to apply the discipline knowledge and capabilities acquired during the programme. A capstone project may incorporate work-integrated learning placements, case analysis and other immersive learning experiences such as study tools, service learning, volunteering, virtual simulations, etc.

Course Weightages:

ATT	CA	MTP	ETP1	ETP2
0	50	0	50	0

Attendance Requirement: NA

Continuous Assessment (CA): NA

CA Evaluation Parameters:

Parameter	Marks
Attendance of the student (* If Att% 76-100 Award:4-5 Marks , 51-75% Award:3 Marks, 26-50% Award:2 Marks, 0-25% Award: 1 Mark)	5
Plan of work was well defined and was realistic to be achieved in stipulated timelines	2
Student has done feasibility analysis required for completion of Project Work such as cost issues, time, operational feasibility etc.	2
Student was able to narrow down the title based upon the preliminary survey.	4

Assistant Dean
Division of Academic Affairs
Lovely Professional University

Testing of the project has been successfully done by the student	3
The categorization of chapters was appropriate and relevant	2
The flow of the written report was well organized and structured.	2
The plagiarism of the report was as per the prescribed limit.	2
The project has been completed and is in presentable form	3
The relevant risks related to the project have been considered.	1
The report was well written and was free from grammatical errors	2
The results obtained by the student are encouraging and supports the proposed work	4
The steps/ action plan/ blueprint/ algorithm/ Methodology formulated by the student is appropriate to solve the problem	4
The student has completed the real implementation of the project work	4
The student has explored some project topics relevant to his/her field/area.	2
The student has explored some real life problems related to which project topic can be finalized.	3
The student has gathered all the relevant tools/ resources/ material required for completion of the project in stipulated timelines	3
The student has identified all the resources/tools/ equipments/ material required for completion of project work till date.	2
The student has started the real implementation of the project work	3
The student has suffcient knowledge regarding the tools/equipments to be used in the project.	3
The student was able to overcome/ handle the problems/ challenges encountere during implementation	3
The title was clear and comprehensive rather than broad and general	3
Title proposed by the student justifies the scope of the proposed work	3
Total	**65**

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CA Evaluation Parameters:

Parameter	Marks
Project Understanding	20
Communication Skills	10
Project Relevance	10
Quality	10
Question Response	20
Skill Set	20
Written Report	10
Total	**100**

Applicability of R Grade: Yes

The conduct of the course shall be as per guidelines issued by the University for the given course.

Assistant Dean
Division of Academic Affairs
Lovely Professional University

CSE493: LINUX SYSTEM ADMINISTRATION

L: 3 T: 0 P: 2 Credits: 4

COURSE OUTCOMES: Through this course students should be able to

CO1 :: recall Linux commands to understand the power of command line to get help, manage files from the command line

CO2 :: apply commands and discuss the creation of users, groups to access files with linux file system permissions

CO3 :: analyze and explain the installation, updation and deletion of software packages

CO4 :: advertise the process priority and tune system performance

CO5 :: demonstrate compressing and deduplicating the storage and determine network - attached storage using the NFS protocol

CO6 :: analyze the boot process to control services offered and to troubleshoot and repair problems

UNIT-I

Accessing the command line : Accessing command line using local console, Accessing command line using desktop, Executing commands using bash shell

Managing Files From the Command Line : The Linux File System Hierarchy, Locating Files by Name, Managing Files Using Command-Line Tools, Matching File Names Using Path Name Expansion, Managing Files with Shell Expansion

Getting Help in Red Hat Enterprise Linux : Reading Documentation Using man Command, Reading Documentation Using pinfo Command, Reading Documentation in /usr/share/doc, Getting Help From Red Hat

Creating, Viewing, and Editing Text Files : Redirecting Output to a File or Program, Editing Text Files from the Shell Prompt, Editing Text Files with a Graphical Editor

Managing Local Linux Users and Groups : Users and Groups, Gaining Superuser Access, Managing Local User Accounts, Managing Local Group Accounts, Managing User Passwords

UNIT-II

Controlling Access to Files with Linux File System Permissions : Linux File System Permissions, Managing File System Permissions from the Command Line, Managing Default Permissions and File Access

Monitoring and Managing Linux Processes : Processes, Controlling Jobs, Killing Processes, Monitoring Process Activity

Controlling Services and Daemons : Identifying Automatically Started System Processes, Controlling System Services

Configuring and Securing OpenSSH Service : Accessing the Remote Command Line with SSH, Configuring SSH Key-based Authentication, Customizing SSH Service Configuration

Analyzing and Storing Logs : System Log Architecture, Reviewing Sys log Files, Reviewing system Journal Entries, Preserving the systemd Journal, Maintaining Accurate Time

UNIT-III

Managing Red Hat Enterprise Linux Networking : Networking Concepts, Validating Network Configuration, Configuring Networking with nmcli, Editing Network Configuration Files, Configuring Host Names and Name Resolution

Archiving and Copying Files Between Systems : Managing Compressed tar Archives, Copying Files Between Systems Securely, Synchronizing Files Between Systems Securely

Installing and Updating Software Packages : Managing Software Updates with yum, Enabling yum Software Repositories, Examining RPM Package Files, Attaching Systems to Subscriptions for Software Updates, RPM Software Packages and Yum

Accessing Linux File Systems : Identifying File Systems and Devices, Mounting and Unmounting File Systems, Making Links Between Files Locating Files on the System

Using Virtualized Systems : Managing a Local Virtualization Host, Installing a New Virtual Machine

UNIT-IV

Improving Command-line productivity : Writing Simple Bash Scripts, Running commands More Efficiently Using Loops, Matching Text in Command Output with Regular Expressions

Scheduling Future Tasks : Scheduling a Deferred User Job, Scheduling Recurring System Jobs, Managing Temporary Files

Tuning System Performance : Adjusting Tuning Profiles, Influencing Process Scheduling

Controlling Access to Files with ACLs : Interpreting File ACLs, Securing Files with ACLs

UNIT-V

Managing SELinux Security : Changing the SELinux Enforcement Mode, Controlling SELinux File Contexts, Controlling SELinux File Contexts, Adjusting SELinux Policy with Booleans, Investing and Resolving SELinux Issues

Managing Basic Storage and Logical Volumes : Adding Partitions, File Systems, and Persistent Mounts, Managing Swap Space, Creating Logical Volumes, Extended Logical Volumes

Implementing Advanced Storage Features : Managing Layered Storage with Stratis, Compressing and Deduplicating Storage with VDO

UNIT-VI

Accessing Network-Attached Storage : Mounting Network-Attached Storage with NFS, Automounting Network-Attached Storage

Controlling the Boot Process : Selecting the Boot Target, Resetting the Root Password, Repairing File System Issues at Boot

Managing Network Security : Managing Server Firewalls, Controlling SeLinux Port Labeling

Installing Red Hat Enterprise Linux : Installing RedHat EnterpriseLinux, Automating Installation with Kickstart, Installing and Configuring Virtual Machines

LABORATORY WORK:

List of Practical's

- Executing commands using bash shell
- Locating Files by Name
- Editing Text Files from the Shell Prompt
- Managing Local User Accounts and Group Accounts
- Managing File System Permissions from the Command Line

Assistant Dean
Division of Academic Affairs
Lovely Professional University

* Killing and Monitoring Process Activity
* Configuring SSH Key-based Authentication
* Validating Network Configuration and Configuring Networking with nmcli
* Installing a New Virtual Machine
* Matching Text in Command Output with Regular Expressions
* Interpreting and Securing Files with ACLs
* Compressing and Deduplicating Storage with VDO
* Mounting Network-Attached Storage with NFS
* Managing Server Firewalls and Controlling SeLinux Port Labeling

RECOMMENDED READINGS:

1. RHCSA & RHCE RED HAT ENTERPRISE LINUX 7: TRAINING AND EXAM PREPARATION GUIDE (EX200 AND EX300) BY ASGHAR GHORI, LIGHTNING SOURCE INCORPORATED

Assistant Dean
Division of Academic Affairs
Lovely Professional University